Ver	Eff_Date	ECN No.	Author	Change Description
1.9_1	08-09-18	E070201828029	James Chang (MPDS)	See Appendix A.11
1.9	04-10-15	E070201507014	Y. H. Weng (MPDS)	See Appendix A.10
1.8	10-24-12	E070201236019	Y. H. Weng (MPDS)	See Appendix A.9
1.7	11-09-11	E070201145027	Y. H. Weng (MDRMP)	See Appendix A.8
1.6	10-26-10	E070201042013	Y. H. Weng (MDRMP)	See Appendix A.7
1.5	01-15-10	E070201002031	Y. M. Chen (PDS)	See Appendix A.6
1.4	10-22-07	E070200734024	D. C. Hu	See Appendix A.5
1.3	06-15-06	E070200620030	R. G. Wu	See Appendix A.4
1.2	06-06-05	E070200521022	Alex Fan	See Appendix A.3
1.1	07-07-04	E070200425024	Alex Fan	See Appendix A.2
1.0	01-02-04	70350035	Alex Fan	Original

Approvals :

Please refer EDW workflow to see detail approval records

Title

TSMC WIRE BOND, FLIP CHIP AND INTERCONNECTION DESIGN RULE (C025-C015)

Document No. : T-000-CL-DR-002

Contents : 70
Attach. : 0
Total : 70

TSMC WIRE BOND, FLIP CHIP AND INTERCONNECTION DESIGN RULE (C025-C015)

Table of Contents

1 Introduction

This chapter contents the following topics:

1.1 Overview

1.2 Applicable Technologies for This Document

1.3 Non-shrinkage Recommendations for Half Node Technologies (C016)

1.4 Instructions on Using This Document

1.1 Overview

This document provides all the rules and reference information for the design and layout of *wire bond* pad packages, the solder *bump pad* of the flip chip package, and redistribution *interconnection*.

The pad pitch and packaging technology should be taken into account at the very first steps of any design project. Special care must be taken regarding power dissipation, functional testability, and assembly.

The *wire bond pad rule* in this document is verified by TSMC's turn key service (probing/assembly) so that probing, bonding, and assembly can be done at most mainstream third-party packaging vendors.

The bump pad rule in this document is verified with the TSMC bump process. If you don't use TSMC bump service, please collect the bump rules from your bump house. Then consult with TSMC before tape out of your chip.

Although each third party's process and capability varies, please consult with your vendor about their own design rules. TSMC works with customers to resolve any design rule conflicts between the probing/bonding/bumping/assembly house and TSMC's fabrication.

1.2 Applicable Technologies for This Document

- This document supports the wire-bond and flip-chip pad solution for TSMC C025/C022/C018/C015 and the related half node (C016) technologies.
- There are 2 kinds of bond package offer in this document, wire bond and flip chip.
- In wire bond pad package, there are 3 kinds of PAD layout (Single-in line, Staggered, Tri-tier) and 2 type of PAD model (Circuit Under Pad, Fully Stacking Pad) make a choice by user.
- 3 kinds of circuit under pad structures structure are offered: {(1)Triple Line (2)One Big Via (3)Single Metal}, and 2 types of metal wire: {(1) Au wire, (2) Cu wire} applied to CUP structure for bond. Offering conditions please look up Table 1.2.2
- In view of the variety of top metal thickness among different technologies, the rules listed in chapter 3 and chapter 4 can support the different top metal thickness in the following table:

Table 1.2.1 The top metal layer and thickness of the applicable technologies in this document

	Technology		Top metal	Thickness (Å)
Al process	**C025/C022**	Logic	M5	8K
	C025	Mixed –mode#1	M5T#2	15K/30K
	C018	Logic	M6	8K
		Mixed -mode	UTM	15K#3/20K/30K#3/40K
	C016	Logic	M6	8K
		Mixed -mode	UTM	20K/40K
	C015	Logic	M7	8K
		Mixed -mode	UTM	30K

#1 Mixed mode is not supported in C022.

#2 M5T and UTM are applied for inductor design.

For the mixed-mode without inductor, the top metal layer and thickness is the same with the Logic process.

#3 C018 15K and 30K UTM is only offered in High Voltage application.

Table 1.2.2 The brief table of top metal layer and thickness of Circuit Under PAD (CUP) structure and wire type choice in this document

Technology	Top Metal THK	CUP	Au Wire	Cu Wire	CP probing
C025	8K	TrL/OBV	o	o	55um*6times (85C)
	15K	TrL/OBV	o	o	75um*6times (85C)
	30K#2	TrL/OBV/SM	o	o	75um*6times (85C)
C022#1	8K	x	x	x	x
C018	8K	TrL/OBV	o	o#4	55um*6times (85C)
	15K#3	TrL	o	o	75um*6times (85C)
	20K	TrL/SM	o	o	75um*6times (85C)
	30K#3	TrL/SM	o	o	75um*6times (85C)
	40K	TrL/SM	o	o	75um*6times (85C)
C016	8K	TrL/OBV	o	o	55um*6times (85C)
	20K	TrL/SM	o	o	75um*6times (85C)
C0152	8K	TrL/OBV	o	o	x
C015	8K	TrL/OBV	o	o	x
	30K#3	TrL/SM	o	o	75um*6times (85C)

#1. Mixed mode is not supported in C022, therefore it doesn't offer UTM
#2. C025 30K UTM is only offered in High Voltage application.
#3. C018 15K and 30K UTM is only offered in High Voltage application.
#4. C018 8K Cu wire recommends the use of the OBV CUP structure.
 # Single Metal abbreviates SM, Triple Line abbreviates TrL, One Big Via abbreviates OBV.
 # The detail condition of qualification as above table, please refer section 3.1.1.
 # x means no offer.

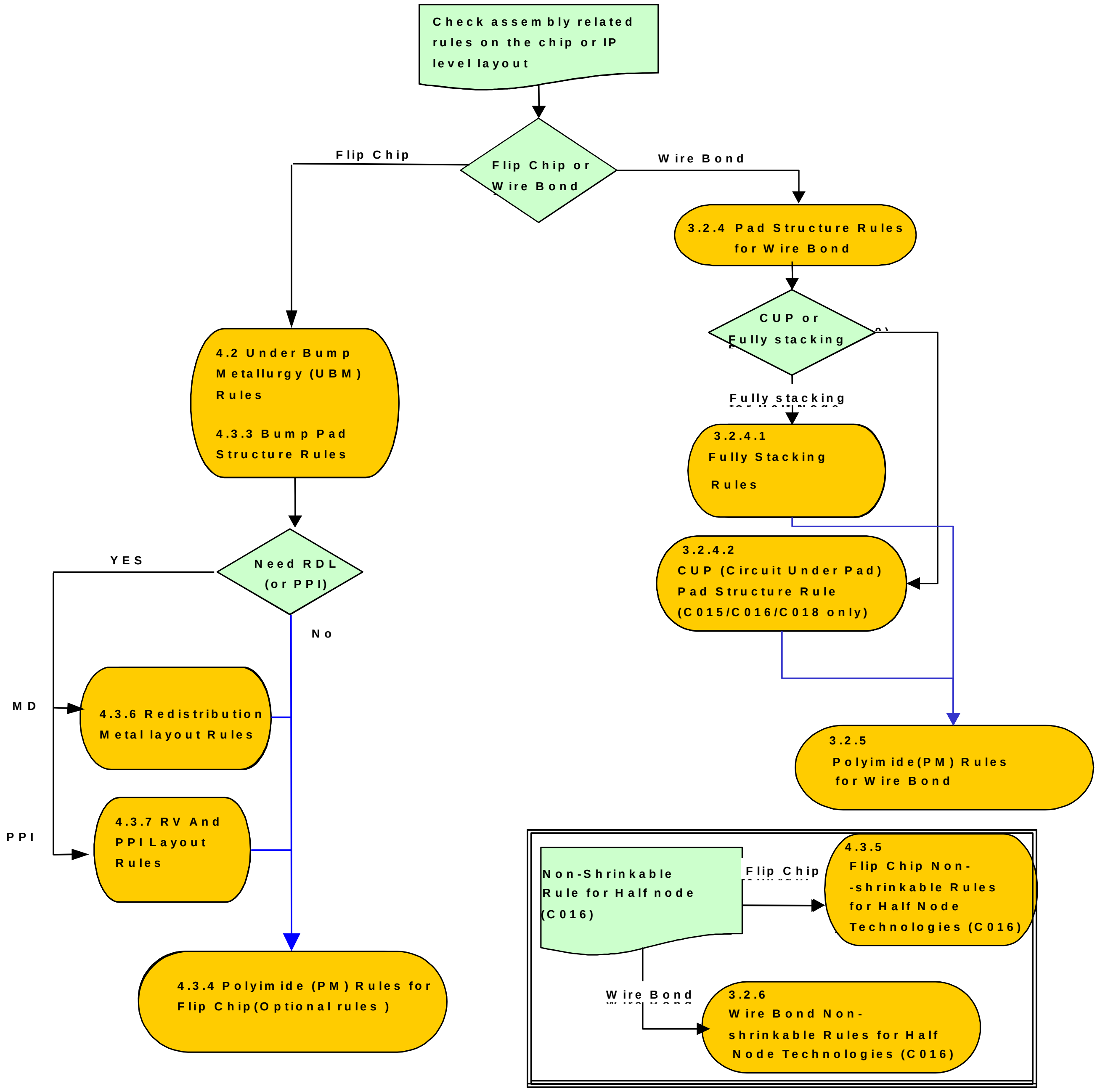
1.3 Non-shrinkage Recommendations for the Half Node Technologies (C016)

1. TSMC also offers 90% shrinkage technologies for die cost saving benefits. For example, C018 shrink to C016.
2. The customer must follow the non-shrinkable rules when using the shrinkage technologies. Please replace the relative dimensions with the non-shrinkable rules in section 3.2.6 and 4.3.5.
3. Besides non-shrinkable rules, other rules allow 90% shrinkage.

1.4 Instructions on Using This Document

This section illustrates the detailed design rules needed for different package options (wire bond or flip chip). Users should follow the instructions to check the related design rules.

Figure 1.4.1 Flow Chart of Design Instructions



2 Layout Rule Conventions, Geometrical Terminology, and Recommendations

2.1 Layout Rule Conventions

Layout rules follow these conventions:

- **Basic units of measure**: The basic unit of measure is μm; the basic unit of area is μm^2.
- **Recommended layout rules**: These rules are designated by a registered symbol ® after the rule number.
- **Guidelines:** These guidelines are designed by a registered symbol "g" after the guideline number.
- Items denoted by "#" depend on the capability of each individual probing/assembly house.
- Items denoted by "^{u}" cannot be checked by DRC.
- **Bracket usage in the rules** should be noted carefully:
  - Parentheses ( ) are used for explanations.
  - Square brackets [ ] are used for certain conditions.
  - Curved brackets { } are used to indicate than an operation is performed.

2.2 Specific Geometries Used in Physical Design Rules

The following table provides definitions for terms that are used in the physical design rules.

Table 2.2.1 Specific Geometries

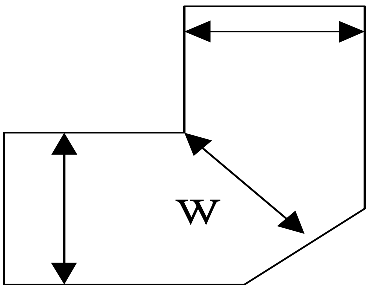
Term	**Definition**
Mtop	The last metal layer, except MD.
Mtop-1	1st metal layer below Mtop
ML	Metal layer with the same metal thickness and design rule as MT Mtop(e.g.: MT Mtop and Mtop-1 of 2 thick metal layer process)
Mx	Metal layer with thin or medium metal thickness (metal layers exclude ML)
Mn	Thick metal layer used on C018
MD	Metal layer of routing redistribution metal layer (RDL)
VIAtop	VIA hole with large VIA size between Mtop and Mtop-1
VIAtop-1	1st VIA hole below VIAtop
VIAL	VIA hole with the same large VIA size and design rule as VIAtop (e.g.: VIAtop and VIAtop-1 of 2 thick top metal layer process)
VIAx	Inter VIA hole is used on C018
VIAn	Top VIA hole used on C018
VIAD	VIA hole between Mtop and MD layer for flip chip
CB	Passivation opening for wire bond design
CBD	Passivation opening for flip chip design
RV	Redistribution VIA for connecting PPI with Mtop
PM	Polyimide opening
PPI	Post passivation interconnection for flip chip
UBM	Under bump metallurgy

Table 2.2.2 Specific Definitions

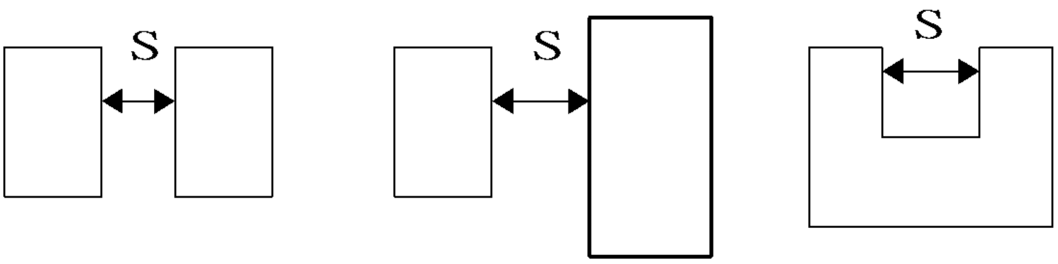
PAD	Metal pad region, {(CB SIZING 2) SIZING -2}
CUP	Wire bond pad design for Circuit Under Pad
FSP	Wire bond pad design for Fully Stacking Pad
DFSP	Wire bond pad design for Deformed Fully Stacking Pad
TrL	Triple Line abbreviates. One kind of via structure.
OBV	One Big Via abbreviates. One kind of via structure.
SM	Single Metal abbreviates.
Chip edge	"Chip" doesn't include seal ring and assembly isolation
Sealring region	10um region with ring type CB from Gds window edge. If you have sealring inside Gds, DRC will use (CB ring in (CHIP NOT (CHIP SIZING -10))) to check 10um sealring region
Assembly isolation	The region between the seal ring and the chip edge

2.3 Layout Geometrical Terminology

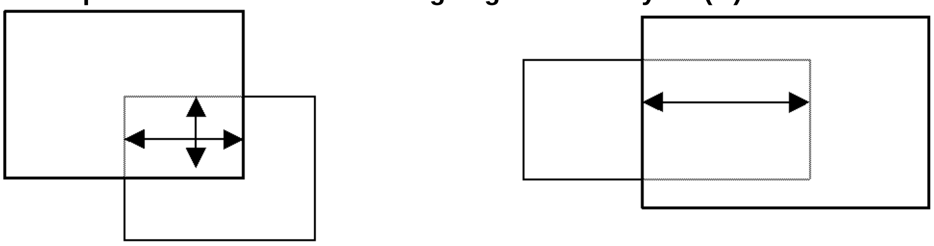
Width: Distance of interior-facing edge for single layer. (W)



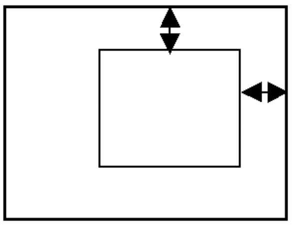
Space: Distance of exterior-facing edge for one or two layer (S)



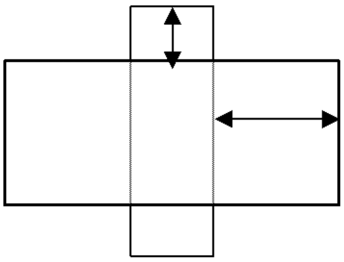
Overlap: Distance of interior-facing edge for two layers (O)



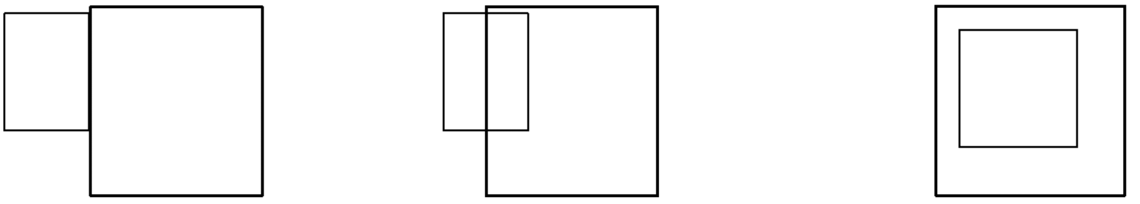
Enclosure: Distance of inside edge to outside edge (Fully inside) (EN)



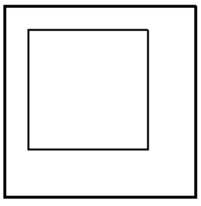
Extension: Distance of inside edge to outside edge (EX)



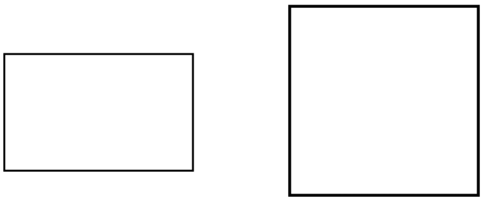
Interact with:



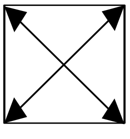
Inside:



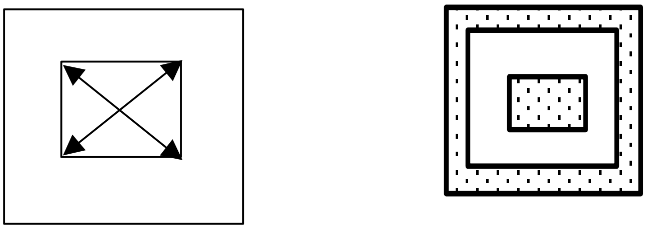
Outside:



Area (A):



Enclosed Area (A):



Pitch (P) = width (W) + space (S)

2.4 Special Recognition CAD Layer Summary

Table 2.4.1 lists special layers used in this design rule and DRC command files. These layers are used for CAD layer recognition and DRC waive. If you use the CAD layers for other applications, be sure that you change the layer mapping in the related deck to make sure the deck can work correctly. Some CAD layer designators include a GDS data-type according to the GDS layer; data-type format.

The column “Tape-out required layers” indicates that this layer must be noted on the mask tape-out form to provide information for mask making.

Table 2.4.1 Special Recognition CAD Layers Summary

Special Layer Name	TSMC Default CAD Layer	Description	Associated With	DRC	Tape-out required layers
DMEXCL	150;x	Dummy Mx exclusion within oxide slot of MT region	DMx rules	✓	
WBDMY	183;0	Design rule waiver within CUP pad region	CUP rules	✓	
RV	-	Redistribution VIA for connecting PPI with Mtop	RV and AP-MD rules	✓	✓
CB	43;0	Passivation opening for wire bond pad	Wire-bond rules	✓	✓
CBD	169;0	Passivation opening for flip chip pad	Flip-chip rules	✓	✓
LMARK	109;0	Cover the laser repair alignment mark region	Wire-bond/ flip-chip rules	✓	✓

Warning: Please use correct CAD layers for each technology by aligning with the associated technology file.

3 Layout Rules for Wire Bond

Need to perform package qualification for change in wafer technology, wafer passivation scheme, die size, package type, package size and package material.

3.1 Recommendations for Wire Bond

3.1.1 General Layout Recommendations

1. **Pad geometry selection:**
   a. Three wire bond geometries are provided: (1) Single in-line; (2) Staggered; and (3) Tri-tiers.
   b. Two wire type solutions are offered in CUP: (1) Au wire; (2) Cu wire. The detail information please consults with TSMC.
   c. Since the variable IO cells might be used on the same chip, different pad geometries on one chip are also allowed. In addition, to meet the pad geometry rules, a space between different pad geometries is also needed. Please refer to the rule of **CB.S.3**.
2. **Pad pitch and width:**
   a. Whenever possible, use a larger pad pitch and size.
   b. It's recommended to use a single pad size on one chip. If you use different pad sizes on one chip, you have to optimize the bonding conditions. You should consult your assembly house carefully.
3. **Polyimide (PM, mask 009):** By default, the PM mask is a reverse tone of the CB mask (or derived from the CB layer by logical operation). Polyimide is required for large chip sizes (> 60 mm^2 or > 100 mm^2 for low stress material packages, for example, BGA). The PM mask can be applied to all types of plastic packages.
4. **Dummy wire bond pads:**
   a. **QFP packages:** Please add two dummy bond pads on each chip corner to form the double bond pads (Fig 3.2.3.1) for wire sweep protection during mold encapsulation of packaging.
   b. If there is space > 150x150(μm) remaining on each chip corner with two dummy pads added, more dummy pads are required to make the space "≤ 150 x 150 (μm)." (Please refer to the "Stress Relief Rule" section in each logic design rule manual.) The pad structure must be comprised of all metal and inter-metal via layers, top metal/…/M2/VIA1/M1**.**
5. **Lead-free packaging with Polyimide coated wafers:** For the lead-free reliability test requirement (260°C reflow temperature of pre-condition test), it's recommended to use a "Green" grade molding compound to prevent PM/molding compound interface delamination.

6. **Test guideline** "Minimum power bus pin numbers per power design is need if Iop>1000mA. " to improve over killed issues and throughput in testing. Minimum power pin number design is as below (M is the power pin number):

$$M_{min} = \frac{10 * I_{op}}{VDD_{nom} - VDD_{min}}$$

where $VDD_{nom} = \frac{VDD_{max} + VDD_{min}}{2}$

VDD_{max} : the maximum allowable operating VDD

VDD_{min} : the minimum allowable operating VDD

I_{op} : the current consumption of a nominal rating power supply VDD

7. The probing condition of CUP qualification

- Top metal = 8 KA, probing overdrive* 55um, touch down 6 times, testing environmental temperature is 85℃*.
- Top metal >= 15 KA, probing overdrive* 75um, touch down 6 times, testing environmental temperature is 85℃*.
- Cantilever probe card specification: Average BCF1.2~1.8 g/mil, Max BCF 2.5 g/mil.

* : TSMC set the worst environmental temperature is 85℃.

* : Overdrive means probe depth. The surface as the origin probes downward certain depth.

8. The condition of wire diameter qualification

- Top metal = 8 KA, Au wire diameter 0.9mil, Cu wire diameter 0.8mil.
- Top metal >= 15 KA, Au wire diameter1.0mil, Cu wire diameter 0.9mil.

3.1.2 Recommendations for Circuit Under Pad (CUP)

1. **Circuit under pad (Abbreviation: CUP):** Metal routing, resistor, de-coupling capacitor, diode, well pickup, IO or ESD circuit are allowed to be placed under pad. However, analog circuit, SRAM, and sensitive circuits should not be placed under the CUP structure. Customers must consult TSMC if other devices are required to be placed under pad.

- Official offer 3 type CUP structure: (1) Single Metal (SM); (2) Triple Line (TrL); (3) One Big Via (OBV). Please refer to below condition to fix your design.
  - If top metal = 8KA, suggest to use OBV CUP.
  - If top metal >= 15KA, suggest to use TrL CUP.
  - If 1P2M scheme and top metal >= 20K, suggest to use SM CUP, [0.25um Gen-2: T-025-CV-DR-015 ; 0.18um Gen-2: T-018-CV-DR-027]

Special metal scheme 1P2M	C025	C018	C016
Fully Stacking	○	○	○
OBV	×	○	○
TrL	○	○	○

2. **Optimizing the wire bonding parameters on the first CUP product:** The bonding condition of CUP is different from the traditional fully stacked pad (non-CUP). It's recommended that the user optimize the bonding conditions on the first CUP product by using the following steps:
   a. Study the bonding conditions first. Use DOE (Design Of Experiment) to optimize the bonding parameters (golden wire, time, power, bonding force, …), and use the wire pull and ball shear test to find the process window.
   b. IMC (Inter Metallurgic Compound) coverage check and a cratered test are also recommended.
   c. TSMC's bonding parameters base on our certain condition, tool, and process, Customer still must use DOE for their product, if user need to refer those bonding parameters, please consult TSMC.
3. **Control the junction temperature (Tj) of High Temperature Operation Life (HTOL) test: Tj<140˚C.**
4. **It is allowed drawing CUP pads and non-CUP pads on the same chip:**

   Please follow the steps below on your first product:
   a. Use the same pad opening size for CUP and non-CUP pads, or as similar size as possible.
   b. WBDMY must be drawn on CUP pads for DRC purposes.
   c. Optimize the bonding condition by following item 2 above. Both CUP and non-CUP must be taken into consideration.
   d. User only selects one type and the same via structure making good use of CUP or deformed fully stacking pad in the same chip.
5. **For RF pad application:**
   a. The CUP pad design can be used for RF pad application.
   b. No circuits, metal routing, or dummy metal layers are allowed within the pad region as defined by WBDMY.
   c. You should apply DMEXCL (CAD layer: 150;x, x=1~8) in each metal layer under the CUP pad region to prevent any dummy metal from being generated within the RF pad region. The region of DMEXCL is illustrated in Fig.3.3.4.2.1.

6. **For pad layout application:**
   - SM & TrL CUP structures allowed TSMC's offering PAD layout including single-in line, staggered, and tri-tier, but OBV is only allowed in single-in line.
7. **For Redistribution Layer process application:**
   - All top vias structure used to specialized pattern PADs (top via are triple line or one big via) are not allowed in redistribution layer process (call RDL or MD), including CUP itself and deformed fully stacking pad.

8. The fully condition for CUP (Circuit Under Pad) offer and FSP (Fully Stacking Pad) copper bond pad structure:

Table 3.1.2.1 0.25um Family

Technology	Top Metal THK	PAD type	VIA structure	Offering Condition	Au Wire	Cu Wire
C025	8K	CUP	TrL	o	o	x
			OBV	o	o	o
			SM	x	x	x
		Fully Stacking Pad	Diamond via array	o	o	x
			Diamond via array +OBV	o	o	o
	15K	CUP	TrL	o	o	o
			OBV	o	o	o
			SM	x	x	x
		Fully Stacking Pad	Diamond via array	o	o	x
			Diamond via array + TrL	o	o	o
			Diamond via array + OBV	o	o	o
	30K	CUP	TrL	o	o	o
			OBV	o	o	o
			SM	o	o	o
		Fully Stacking Pad	Diamond via array	o	o	x
			Diamond via array + TrL	o	o	o
			Diamond via array + OBV	o	o	o
C022	8K	CUP	x	x	x	x
		Fully Stacking Pad	Diamond via array	o	o	x

1. ○ is offer; × is no offer, diamond via array refer to section 3.2.4.1 Figure, diamond via array + TrL means VIA1 to VIAtop-1 is diamond via array, VIAtop is TrL structure. Diamond via array + OBV means VIA1 to VIAtop-1 is diamond via array, VIAtop is OBV structure. The Structures refer to section 3.2.4.2.1 and section 3.2.4.2.1.
2. There is no Single Metal structure in deformed fully stacking PAD.
3. Fully stacking means metal/via all exist in every layer. CUP is only Mtop/VIAtop/Mtop-1.
4. Deformed fully stacking PAD offering is base on CUP offering in the same metal thickness in the same technology.

Table 3.1.2.2 0.18um Family

Technology	Top Metal THK	PAD type	VIA structure	Offering Condition	Au Wire	Cu Wire
C018	8K	CUP	TrL	o	o	x
			OBV	o	o	o
			SM	x	x	x
		Fully Stacking Pad	Diamond via array	o	o	x
			Diamond via array + OBV	o	o	o
	15K	CUP	TrL	o	o	o
			OBV	x	x	x
			SM	x	x	x
		Fully Stacking Pad	Diamond via array	o	o	x
			Diamond via array + TrL	o	o	o
	20K	CUP	TrL	o	o	o
			OBV	x	x	x
			SM	o	o	o
		Fully Stacking Pad	Diamond via array	o	o	x
			Diamond via array + TrL	o	o	o
	30K	CUP	TrL	o	o	o
			OBV	x	x	x
			SM	o	o	o
		Fully Stacking Pad	Diamond via array	o	o	x
			Diamond via array + TrL	o	o	o
	40K	CUP	TrL	o	o	o
			OBV	x	x	x
			SM	o	o	o
		Fully Stacking Pad	Diamond via array	o	o	x
			Diamond via array + TrL	o	o	o
C016	8K	CUP	TrL	o	o	x
			OBV	o	o	o
			SM	x	x	x
		Fully Stacking Pad	Diamond via array	o	o	x
			Diamond via array + OBV	o	o	o
	20K	CUP	TrL	o	o	o
			OBV	x	x	x
			SM	o	o	o
		Fully Stacking Pad	Diamond via array	o	o	x
			Diamond via array + TrL	o	o	o
C0152	8K	CUP	TrL	o	o	x
			OBV	o	o	o
		Fully Stacking Pad	Diamond via array	o	o	x
			Diamond via array + OBV	o	o	o
C015	8K	CUP	TrL	o	o	x
			OBV	o	o	o
		Fully Stacking Pad	Diamond via array	o	o	x
			Diamond via array + OBV	o	o	o
	30K	CUP	TrL	o	o	o
			SM	o	o	o
		Fully Stacking Pad	Diamond via array	o	o	x
			Diamond via array + TrL	o	o	o

1. ○ is offer; ╳ is no offer, diamond via array refer to section 3.2.4.1 Figure, diamond via array + TrL means VIA1 to VIAtop-1 is diamond via array, VIAtop is TrL structure. Diamond via array + OBV means VIA1 to VIAtop-1 is diamond via array, VIAtop is OBV structure. The Structures refer to section 3.2.4.2.1 and section 3.2.4.2.1.
2. There is no Single Metal structure in deformed fully stacking PAD.
3. Fully stacking means metal/via all exist in every layer.
4. CUP is only Mtop/VIAtop/Mtop-1(Trl / OBV) or Mtop (SM).
5. Deformed fully stacking PAD offering is base on CUP offering in the same metal thickness in the same technology.

3.2 Wire Bond Rules

3.2.1 Mask Information, Process Sequence, and CAD Layers

The table below lists masks and corresponding masking steps.

- The mask sequence in the following tables start from the top metal. Please refer to the DRM of each technology for the mask sequences before the top metal.
- **Mask Name column:** This column lists names that are reserved for standard mask steps. These names should not be used for another purpose without prior authorization from TSMC.
- **CAD Layer column:** This column lists CAD layer numbers.

Warning: A CAD layer number must be ≤ 255. If the CAD layer number is > 255, the mask making will fail.

Table 3.2.1.1 Mask Information and CAD layers (C025/C022/C018/C015)

Sequence	Mask Name	Mask ID	Digitized Area (Dark or Clear)	CAD Layer** (C018/C022/C025 only)	Reference Layer in Logical Operation	Type	Description
						WB	
1	CB	107	C	19	-	**1**	Passivation pad opening for wire bond
2	PM	009	D	Derived	CB, FW, LMARK	**#**	Polyimide window for wire bond

The definition of WB is wire bond and the legends (0, 1, and *) mean:
0 : Does not use the mask
1 : Must use the mask
 # : Polyimide is required for large chip sizes (> 60 mm2 or > 100 mm2 for low stress material packages, for example, BGA)
‡: Follow the mask and CAD layer of the metal/VIA layer above the Mtop/VIAtop.

Warning: Some of the CAD layer numbers for C015 are different from other processes (C018/C022/C025). The CAD layer provided in Table 3.2.1.1 is only for C018/C022/C025. Please align the correct CAD layer for each technology with the associated technology file.

**** CAD layer table (C025/C022/C018/C015):**

Mask Name	Mask ID	Digitized Area (Dark or Clear)	CAD Layer (C015 only)	CAD Layer (C018 only)	CAD Layer (C022 only)	CAD Layer (C025 only)
CB	107	C	43(CB)	19(CB)	19(CB)	19(CB)
PM	009	D	Derived	Derived	Derived	Derived

3.2.2 Reference Document

Content	Reference Documentation
Qualification report	• S-QCS-04-03-005 *POST CP WAFER QUALITY ASSURANCE SPECIFICATION (BUMP & NON-BUMP WAFER)*
	• T-018-CL-QR-013 *TSMC 0.18UM CIRCUIT UNDER PAD 1P6M SALICIDE ALCU 1.8&3.3V EVALUATION REPORT*
	• T-025-CL-QR-014 *TSMC 0.25 UM CMOS LOGIC CUP WIRE BOND PACKAGE QUALIFICATION REPORT*
Wafer sort	• T-018-CL-RP-012 *Wire Bond and Wafer Sort Reference Report for TSMC 0.18UM Circuit Under Pad Design*

3.2.3 Passivation Layer Open Rules

- The 80µm tri-tier pitch is only for the BGA package.

Passivation window (CB)

Rule No.	Description	Label		Rule
Single in-line (Figure 3.2.3.1) / Staggered (Figure 3.2.3.2) / Tri-Tier (Figure 3.2.3.3)				
#CB.W.1	Width of CB	A	≥	Table 3.2.3.1
#CB.W.2	Length of CB (the edge perpendicular to nearby chip edge)	A1	≥	Table 3.2.3.1
CB.W.3	Width of pad metal under Mtop-2 within CB region Staggered: M1, M2~Mtop-3 under inner pad Tri-tier: M1, M2~Mtop-3 under middle pad M2~Mtop-3 under inner pad C015/C018 CUP pad are excluded.	E	≥	Table 3.2.3.1
CB.W.4®u	Recommended total width of all metal layers connecting with bond pad. (Refer to the “I/O ESD Protection Guideline” section in the corresponding design rule manual.)	W	≥	20
#CB.S.1	Space of two CB	B	≥	Table 3.2.3.1
CB.S.2	Space of metal pad geometries, or space of metal pad to metal line DRC check metal pad by the following definitions: (1) {Mx AND [CB sizing 3]}, or (2) {(Mtop/Mtop-1) AND [CB sizing 3]} for C015/C018 CUP pad	F	≥	2.5
#CB.S.3	Space of the different pad geometries [(Single to Stagger), (Single to Tri-tier) or (Stagger to Tri-tier)]	B1	≥	15
#CB.P.1u	Pitch of (4 pitches at each chip corner) Note: Starting from the corners, the first four pitches must be larger than the pitch in the regularly repeating pad center area.	C1	≥	Table 3.2.3.1
#CB.P.2 u	Pitch of x-direction (inner and middle pad of Tri-Tier)	C2	≥ ≤	13 40
#CB.P.3 u	Pitch of x-direction [(inner and outer pad of Staggered) or (middle and outer pad of Tri-Tier)]	C3	≥	Table 3.2.3.1
#CB.P.4 u	Pitch of y-direction (between inner-middle-outer pad) Staggered: inner-outer pad Tri-Tier: middle-outer pad, and inner-middle pad	C4	≥	Table 3.2.3.1
CB.EN.1	CB enclosure by Mx and Mtop in pad regions Except, Staggered: M1, M2~Mtop-3 under inner pad Tri-Tier: M1, M2~Mtop-3 under middle and inner pad C015/C018 CUP pad: M1~Mtop-2	D	≥	2 (Al process)
#CB.R.1	For pad pitch < 55µm, one pair of notches on each pad CB is required, to be the reference point for probing and bonding. Please refer to Figure 3.2.3.1 for the size and location.			
#CBWB.R.1u	WB mark is always to define as far as possible in the middle bounding side BS, if there is any slanting. Please follow as below statement. WB mark enclosure by CB in pad regions: Wire bonding mark area "X"um/per-side than original bonding pad area, because OBV film step high issues. If TM =8K, X> 0.8um, If TM =15K, X> 1.5um, If TM =20K, X> 2um, If TM =40K, X> 4um.	X		

Table 3.2.3.1: Rule Summary of item A~E

Pitch(C)	A	A1	B	C1	C3	C4	E
≥50 single in-line	43	80	7	80			
≥55 single in-line	48	66	7	80			
≥60 staggered	53	66	7	100	30	96	20
≥70 staggered (Recommended)	58	72	12	100	35	102	45
≥80 staggered (Recommended)	65	75	15	108	40	115	50
≥80 tri-tier (BGA only)	65	75	15	108	40	115	50

This pad geometry can allow tighter pitch (<55µm), but it needs the CB notch pairs to separate the probing and bonding locations (CB.R.1).

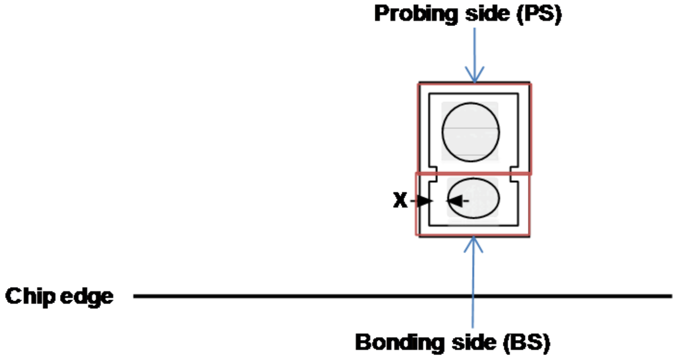


Figure 3.2.3.1: Single in -line

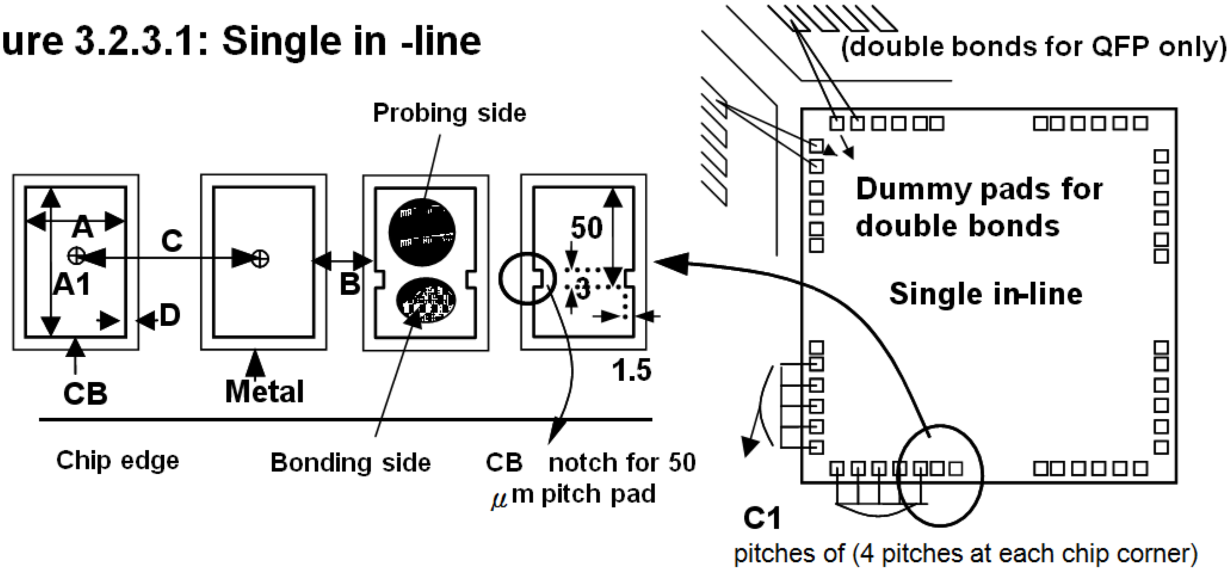


Figure 3.2.3.2: Staggered

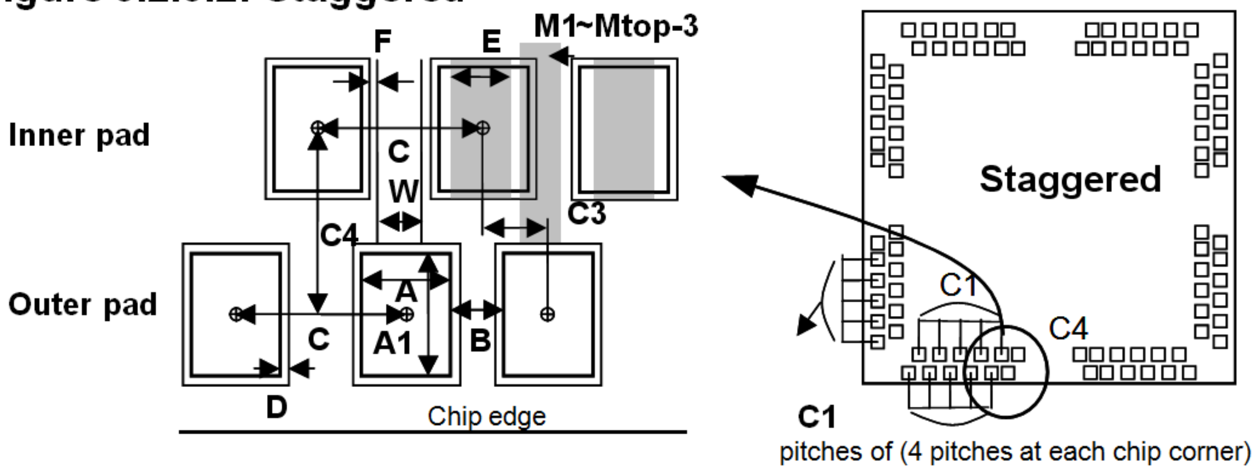
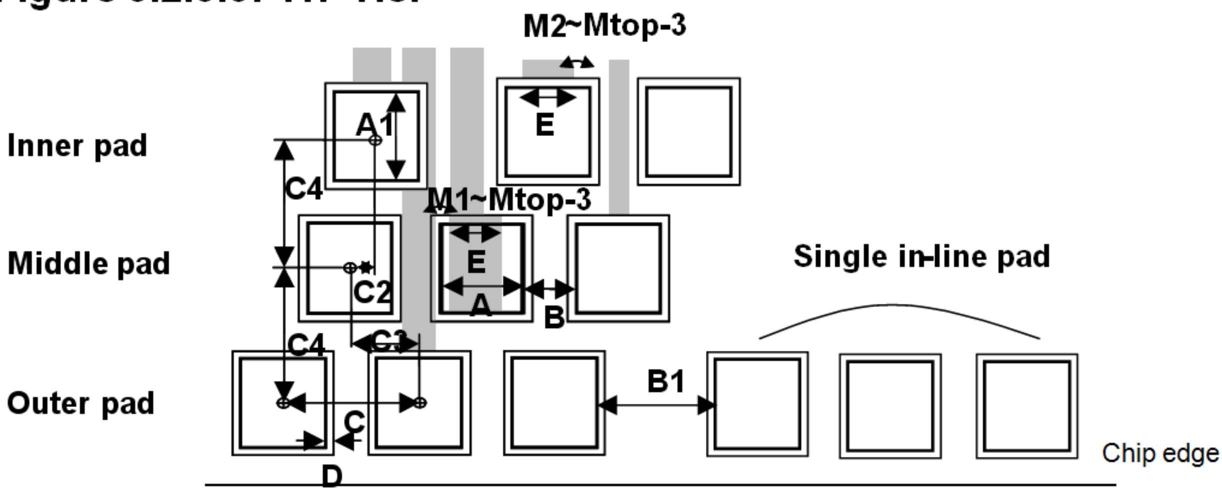


Figure 3.2.3.3: Tri -Tier



3.2.4 Pad Structure Rules (C025~C015)

3.2.4.1 Full Stacking Pad structure rules

Metal/Via/Passivation Window (CB) Rules

Rule No.	Description	Label		Rule
Common items (for Single In-Line/Staggered/Tri-Tier)				
CB.EN.1	Enclosure by Mx and Mtop pad regions Except, Staggered: M1~Mtop-3 under inner pad Tri-Tier: M1~Mtop-3 under middle and inner pad	D	≥	2
CB.R.2	1. Pad structure must be at least Mtop/VIAtop/Mtop-1…VIA1/M1. Exception: inner pad of Tri-Tier structure can be Mtop…VIA2/ M2. M1 routing under pad is allowed. (CUP structure without WBDMY is not allowed. Please make sure that drawing the dummy layer WBDMY is for CUP structure purpose.) 2[U]. Vias should form a diamond pattern in CB center.			
CB.R.3	Slots in metal pad region are not allowed, DRC check metal pad by the following definitions: (1) {Mx AND [CB sizing 3]}, or (2) {(Mtop/Mtop-1) AND [CB sizing 3]}			

Rules G, H, I, and J in the table below have different dimensions for C025~C015 processes. For more than one thick metal layer with top metal thickness, these thick metal layers are named ML. VIA layers below the thick metal are named VIAL.

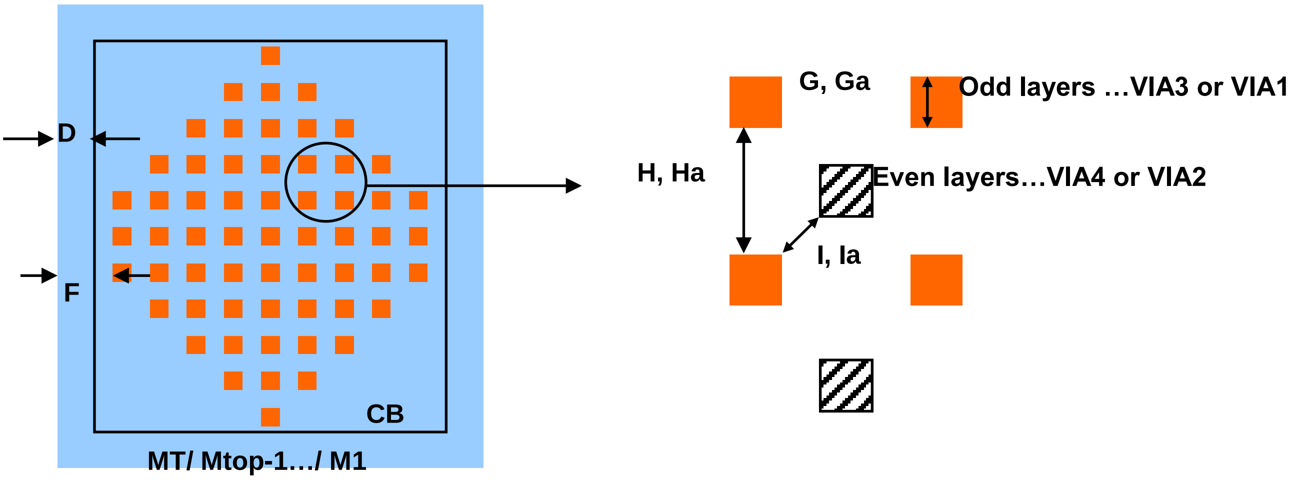
CBVIAx.W.1	Width of VIAx (maximum = minimum)	G	=	Table 3.2.4.1
CBVIAx.S.1	Space of two VIAx	H	≥	Table 3.2.4.1
CBVIAx.S.2	Space to VIAx-1, or to VIAx+1	I	≥	Table 3.2.4.1
CBVIAx.EN.1	VIAx/VIAtop/VIAL enclosure by metal layer [the nearest via on the four corner of diamond] Except: Staggered: M1~Mtop-3 under inner pad Tri-Tier: M1~Mtop-3 under middle and inner pad	F	≥	2
			≤	4
CBVIAx.R.1	Ratio of total exposure area of {VIA hole inside CB} to CB area, except; VIA1, VIA2~VIAtop-2 of the inter row pad for Staggered pad VIA1, VIA2~VIAtop-2 of the both inter/middle row pad of Tri-tier pad	J	≥	Table 3.2.4.1
CBVIAx.R.1.1	Ratio of total exposure area of {VIA1, VIA2~VIAtop-2 inside CB} to CB area (In the inter row pad of staggered and the inter/middle row pad of Tri-tier)	J1	≥	Table 3.2.4.1
CBVIAT.W.1	Width of VIAtop and width of VIAL	Ga	=	Table 3.2.4.1
CBVIAT.S.1	Space of two VIAtop, and space of two VIAL	Ha	≥	Table 3.2.4.1
CBVIAT.S.2	Space to VIAtop-1, and VIAL space to the most upper VIAx under VIAL	Ia	≥	Table 3.2.4.1

Table 3.2.4.1: Rule Summary of Item G~J1

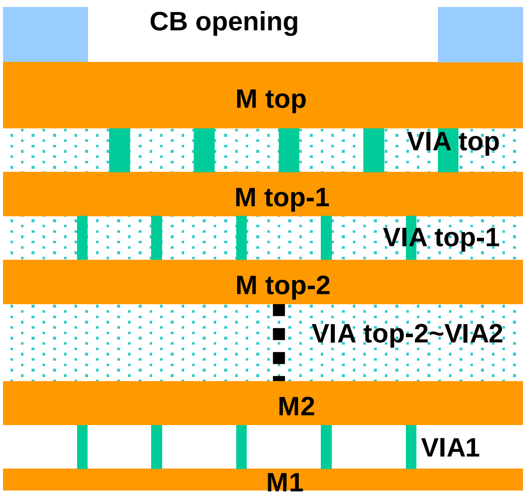
Technology	G	H	I	Ga	Ha	Ia	J	J1
C025	0.36	0.68	0.20				5%	2.5%
C022	0.32	0.68	0.20				5%	2.5%
C018	0.26	0.45	0.13	0.36	0.35	0.06	5%	2.5%
C015	0.22	0.49	0.13	0.36	0.35	0.06	4%	2.5%

Full stacking pad structure with diamond pattern VIA:

Top View:



Cross Section:



3.2.4.2 Deformed Fully Stacking Pad (DFSP) structure rules for copper bond

TSMC also offered the copper bond pad in fully stacking structure, but layout pattern had a little different from originals. If customer required making use of copper wire bond, the pad structures please use the CUP top via alike in place of the fully stacking pad top via (One big via or Triple line structure replaced via array in top via region) please follow as below figure for reference. In other words VIA1 to VIAtop-1 keeping via array and VIAtop is one big via or triple line. The copper bond pad structure cross sections refer to the figure, and which top via type used, illustrated the condition with table 3.1.2.1 and table 3.1.2.2.

3.2.4.2.1 Diamond via array deformation (1) C025/C018/C016/C0152/C015 via array + TrL

Rule No.	Description	Label		Rule
Common items (for Single In-Line/ Staggered/ Tri-Tier)				
DFSCB.EN.1	Enclosure by MT/Mtop-1	D	≥	2
DFSCB.R.1 [u]	VIAtop pattern under CB region must be composed of the vertical and horizontal triple-line cells. Please refer to **Figure 3.2.4.2.1** and **Figure 3.2.4.2.2** for the correct top VIA drawing.			
DFSVIAT.EN.1	Triple-line VIAtop enclosure by Mtop/ Mtop-1	G1	≥	2
DFSVIAT.W.1	Width of line VIAtop	H1	=	0.28
DFSVIAT.W.2	Length of line VIAtop	Ha1	=	3.44
DFSVIAT.S.1	Space of two line VIAtop [in parallel direction] (within the same Triple-line VIA cell)	I1	=	1.3
DFSVIAT.S.2	Space of two line VIAtop [in perpendicular direction] (between two neighboring Triple-line VIA cells)	J1	=	2
DFSVIAT.DN.1	Ratio of total exposure line VIAtop area to CB area within CB region		≥	8%
DFSCB.EN.2	Enclosure by Mtop-1 pad regions Except, Staggered: M1~Mtop-3 under inner pad Tri-Tier: M1~Mtop-3 under middle and inner pad	D1	≥	2
CB.R.2	1. Pad structure must be at least Mtop/VIAtop/Mtop-1…VIA1/M1. Exception: inner pad of Tri-Tier structure can be Mtop…VIA2/ M2. M1 routing under pad is allowed. 2[U]. VIAs should form a diamond pattern in CB center except top via.			
CBVIAx.W.1	Width of VIAx (maximum = minimum)	G	=	Table 3.2.4.2.1
CBVIAx.S.1	Space of two VIAx	H	≥	Table 3.2.4.2.1
CBVIAx.S.2	Space to VIAx-1, or to VIAx+1	I	≥	Table 3.2.4.2.1
CBVIAx.EN.1	VIAx enclosure by metal layer [the nearest via on the four corner of diamond] Except: Staggered: M1~Mtop-3 under inner pad Tri-Tier: M1~Mtop-3 under middle and inner pad	F	≥ ≤	2 4
CBVIAx.R.1	Ratio of total exposure area of {VIA hole inside CB} to CB area, except; VIA1, VIA2~VIAtop-2 of the inter row pad for Staggered pad VIA1, VIA2~VIAtop-2 of the both inter/middle row pad of Tri-tier pad	J	≥	Table 3.2.4.2.1
CBVIAx.R.1.1	Ratio of total exposure area of {VIA1, VIA2~VIAtop-2 inside CB} to CB area (In the inter row pad of staggered and the inter/middle row pad of Tri-tier)	J1	≥	Table 3.2.4.2.1
TRL.R.1	It defined TrL as VIAtop in CUP, you also must use TrL deformed fully stacking PAD if non-CUP demand in whole chip.			

Rule No.	Description	Label		Rule
Common items (for Single In-Line/ Staggered/ Tri-Tier)				
CB.R.3	Slots in metal pad region are not allowed, DRC check metal pad by the following definitions: (1) {Mx AND [CB sizing 3]}, or (2) {(Mtop/Mtop-1) AND [CB sizing 3]}			

Table 3.2.4.2.1: Rule Summary of Item G1~J2

Technology	G	H	I	J	J1
C025	0.36	0.68	0.20	5%	2.5%
C022	0.32	0.68	0.20	5%	2.5%
C018	0.26	0.45	0.13	5%	2.5%
C015	0.22	0.49	0.13	4%	2.5%

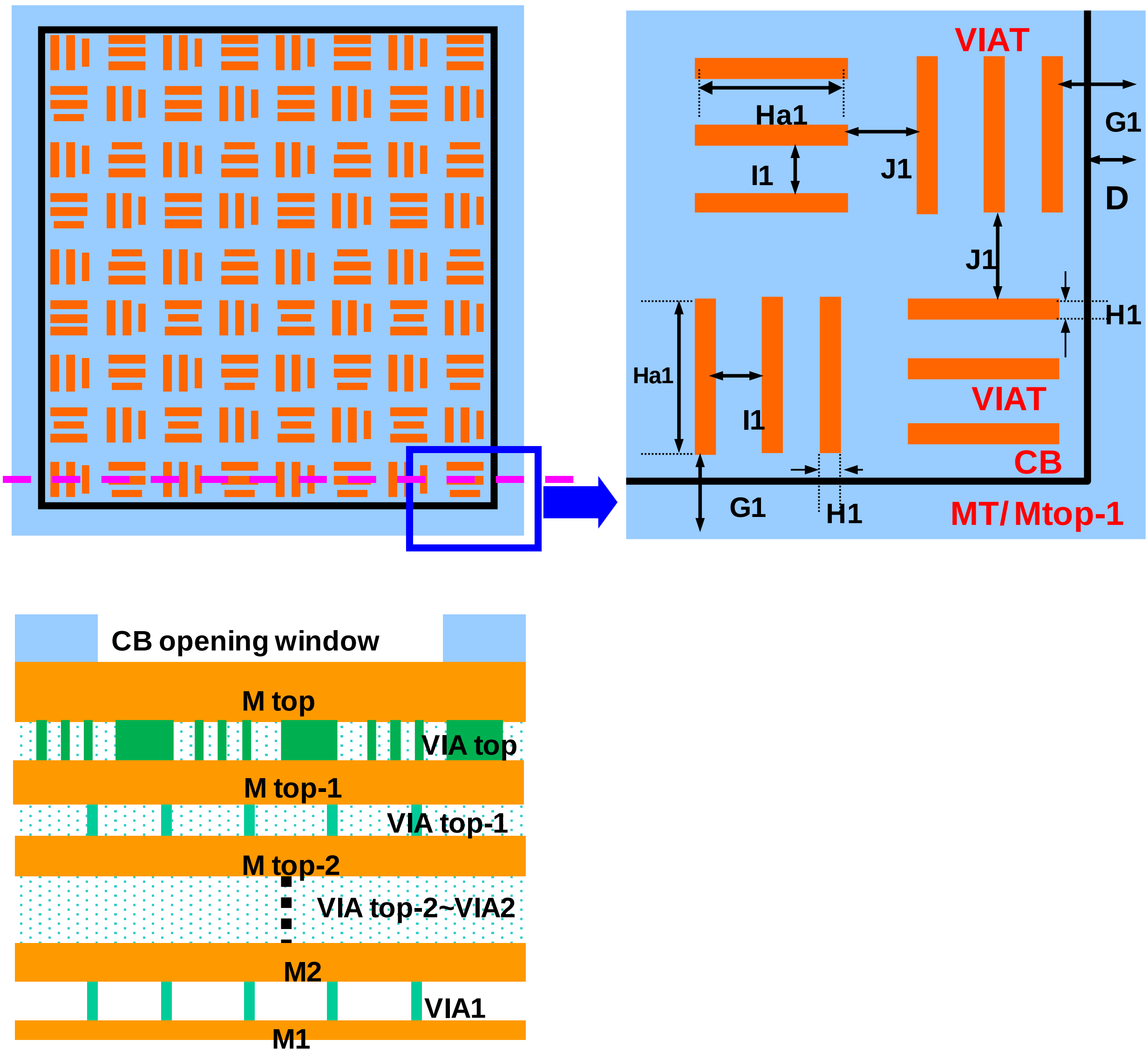


Figure 3.2.4.2.1 The top view, cross section and details of diamond via array + TrL

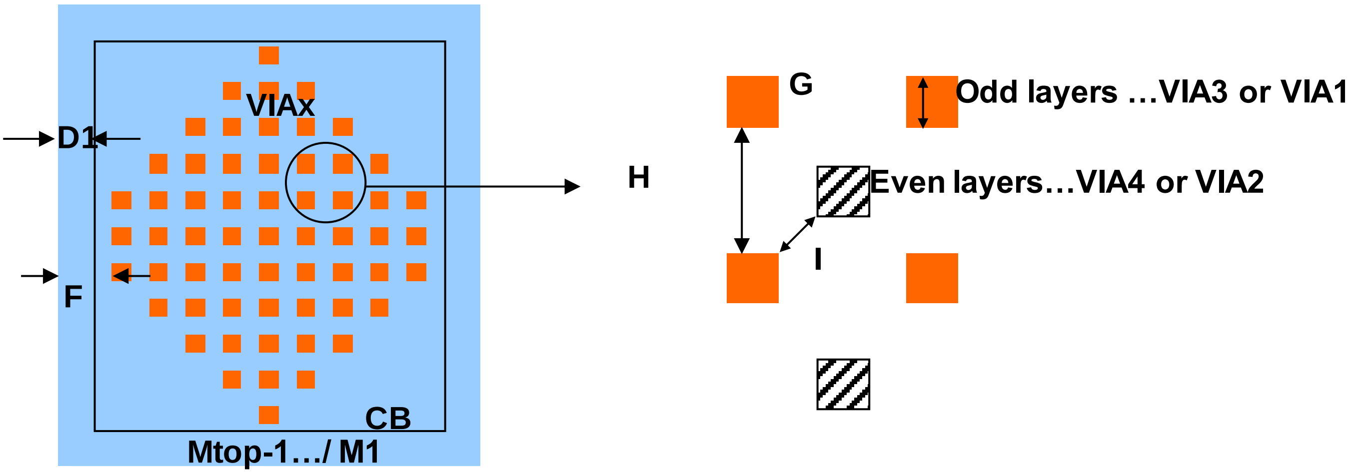


Figure 3.2.4.2.2 Mtop-1/VIAtop-1 and below in DFS Pad

3.2.4.2.2 Diamond via array deformation (2) C025 via array + OBV

Rule No.	Description	Label		Rule
Common items (for Single In-Line)				
DFSCB.EN.3	CB enclosure by Mtop/Mtop-1	D	≧	2
DFSCB.R.2	VIAtop must be exactly identical to {(CB sizing 2) sizing -2} in pad structure (Notch exclusion). Please refer to **Figure 3.2.4.2.3** and **Figure 3.2.4.2.4** for the correct top VIA drawing.			
DFSVIAT.W.3	The maximum width of top via.	Y	≦	300
DFSVIAT.EN.1	VIAtop enclosure by Mtop/ Mtop-1	G	≧	2
DFSVIAT.DN.1	The maximum density of top via (all of top vias) across whole chip		<	10%
DFSOBV.R.1	It defined OBV as VIAtop, you must use OBV CUP PAD if CUP demand in whole chip.			
DFSCB.EN.4	Enclosure by Mtop-1 pad regions.	D1	≥	2
CB.R.2	1. Pad structure must be at least Mtop/VIAtop/Mtop-1…VIA1/M1. 2[U]. VIAs should form a diamond pattern in CB center except top via.			
CBVIAx.W.1	Width of VIAx (maximum = minimum)	G	=	Table 3.2.4.2.1
CBVIAx.S.1	Space of two VIAx	H	≥	Table 3.2.4.2.1
CBVIAx.S.2	Space to VIAx-1, or to VIAx+1	I	≥	Table 3.2.4.2.1
CBVIAx.EN.1	VIAx enclosure by metal layer [the nearest via on the four corner of diamond]	F	≥	2
			≤	4
CBVIAx.R.1	Ratio of total exposure area of {VIA hole inside CB} to CB area.	J	≥	Table 3.2.4.2.1
CBVIAx.R.1.1	Ratio of total exposure area of {VIA1, VIA2~VIAtop-2 inside CB} to CB area.	J1	≥	Table 3.2.4.2.1
OBV.R.2	It only allowed single in line PAD distribution in the whole chip			
CB.R.3	Slots in metal pad region are not allowed, DRC check metal pad by the following definitions: (1) {Mx AND [CB sizing 3]}, or (2) {(Mtop/Mtop-1) AND [CB sizing 3]}			

Table 3.2.4.2.1: Rule Summary of Item G~J1

Technology	G	H	I	J	J1
C025	0.36	0.68	0.20	5%	2.5%
C022	0.32	0.68	0.20	5%	2.5%
C018	0.26	0.45	0.13	5%	2.5%
C015	0.22	0.49	0.13	4%	2.5%

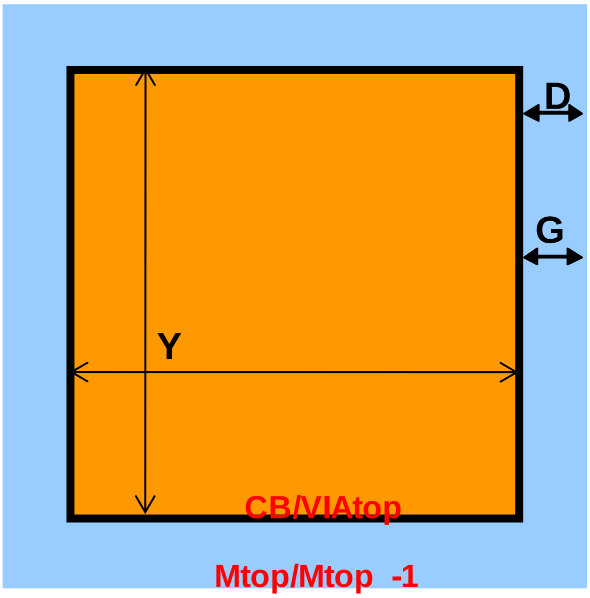


Figure 3.2.4.2.3 One big via top view

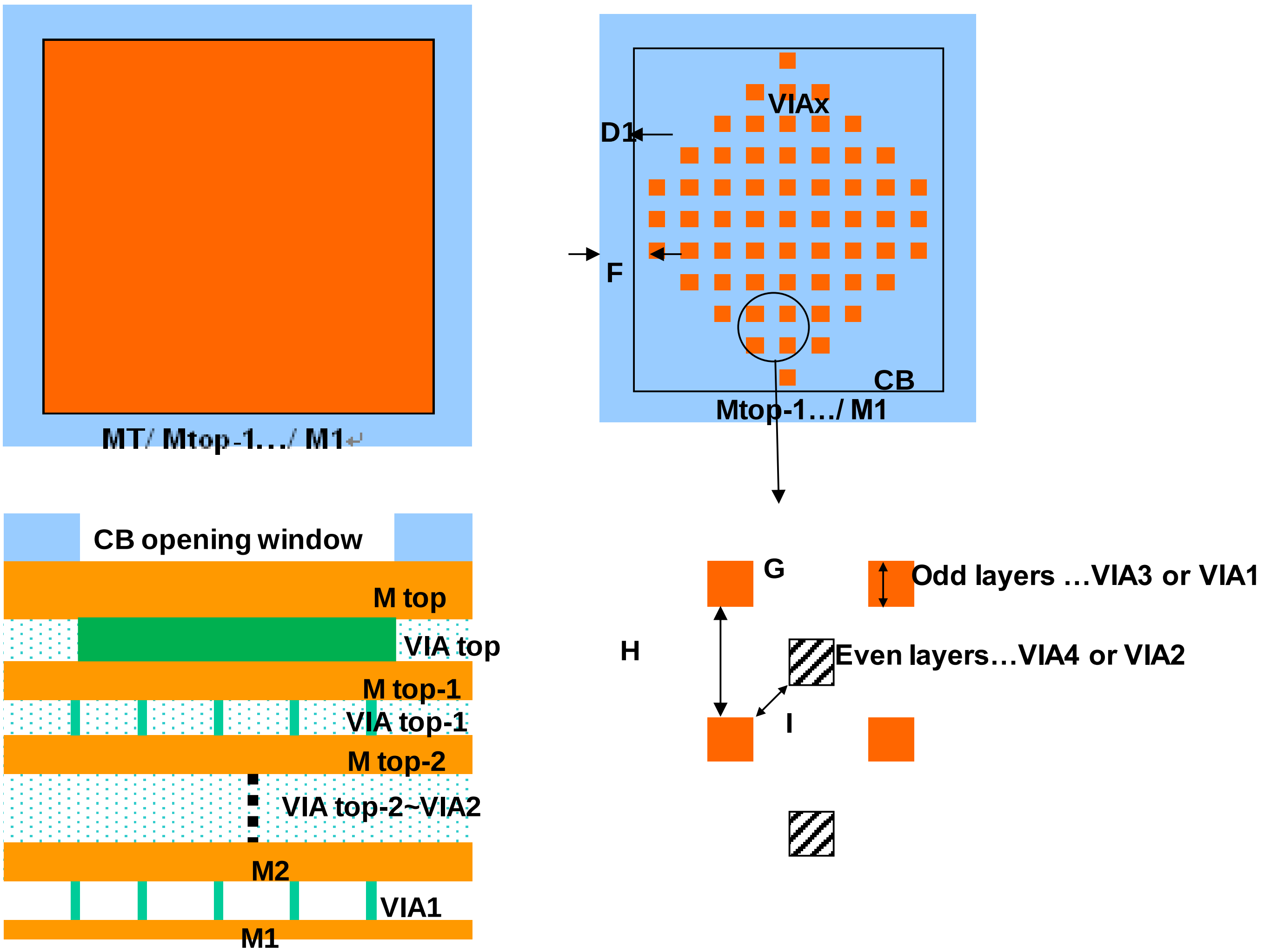


Figure 3.2.4.2.3 The top view, cross section and details of diamond via array + OBV (Left side)
Figure 3.2.4.2.4 Mtop-1/VIAtop-1 and below in DFS Pad (Right side)

3.2.4.2.3 Diamond via array deformation (3) C018/C016/C0152/C015 via array + OBV

Rule No.	Description	Label		Rule
Common items (for Single In-Line)				
DFSCB.EN.5	CB enclosure by Mtop	D	≧	2
DFSCB.R.2	VIAtop must be exactly identical to {(CB sizing 2) sizing -2} in pad structure (Notch exclusion). Please refer to **Figure 3.2.4.2.5** for the correct top VIA drawing.			
DFSVIAT.W.4	The maximum width of top via.	Y	≦	150
DFSVIAT.S.3	The minimum space of one big via to one big via.	X	≥	7
DFSVIAT.EN.2	VIAtop enclosure by Mtop.	G	≧	2
DFSVIAT.EN.3	VIAtop enclosure by Mtop-1 except butted metal interconnection. Metal width in {(CB SIZING 2) NOT (CB SIZING 0.2)} region must be less than the Mtop-1 pad dimension.	G1	=	0.2
DFSVIAT.DN.1	The maximum density of top via (all of top vias) across whole chip		<	10%
DFSOBV.R.1	If defined OBV as VIAtop, you must use OBV CUP PAD if CUP demand in whole chip.			
DFSCB.EN.4	Enclosure by Mtop-2 pad regions	D1	≥	2
CB.R.2	1. Pad structure must be at least Mtop/VIAtop/Mtop-1…VIA1/M1. 2 U. VIAs should form a diamond pattern in CB center except top via.			
CBVIAx.W.1	Width of VIAx (maximum = minimum)	G	=	Table 3.2.4.2.1
CBVIAx.S.1	Space of two VIAx	H	≥	Table 3.2.4.2.1
CBVIAx.S.2	Space to VIAx-1, or to VIAx+1	I	≥	Table 3.2.4.2.1
CBVIAx.EN.1	VIAx enclosure by metal layer [the nearest via on the four corner of diamond]	F	≥	2
			≤	4
CBVIAx.R.1	Ratio of total exposure area of {VIA hole inside CB} to CB area.	J	≥	Table 3.2.4.2.1
CBVIAx.R.1.1	Ratio of total exposure area of {VIA1, VIA2~VIAtop-2 inside CB} to CB area.	J1	≥	Table 3.2.4.2.1
OBV.R.2	It only allowed single in line PAD distribution in the whole chip.			
CB.R.3	Slots in metal pad region are not allowed, DRC check metal pad by the following definitions: (1) {Mx AND [CB sizing 3]}, or (2) {(Mtop/Mtop-1) AND [CB sizing 3]}			

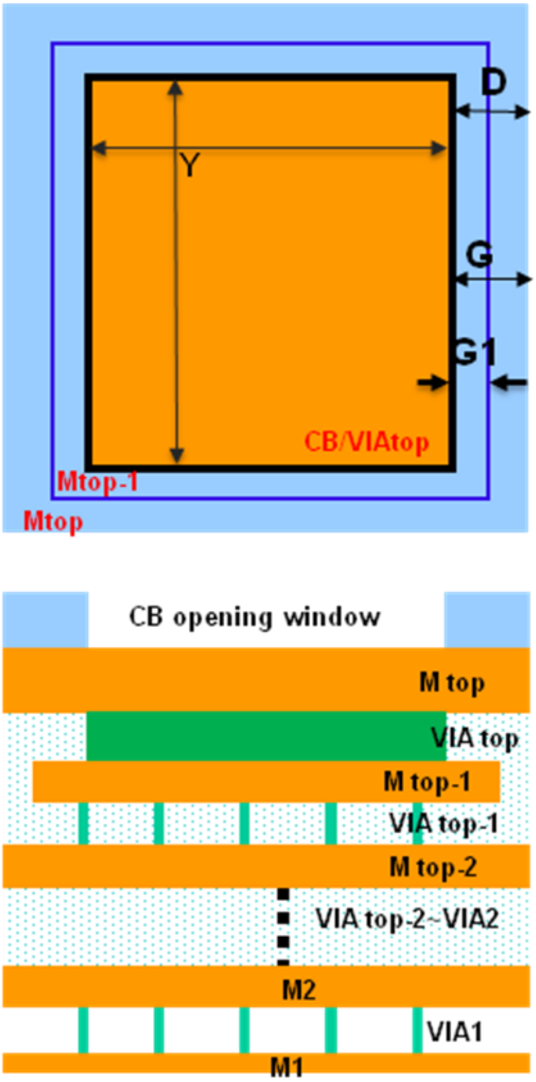


Figure 3.2.4.2.5 The top view and cross section of diamond via array + OBV

3.2.4.3 Triple Line (TrL) CUP Pad Structure Rules

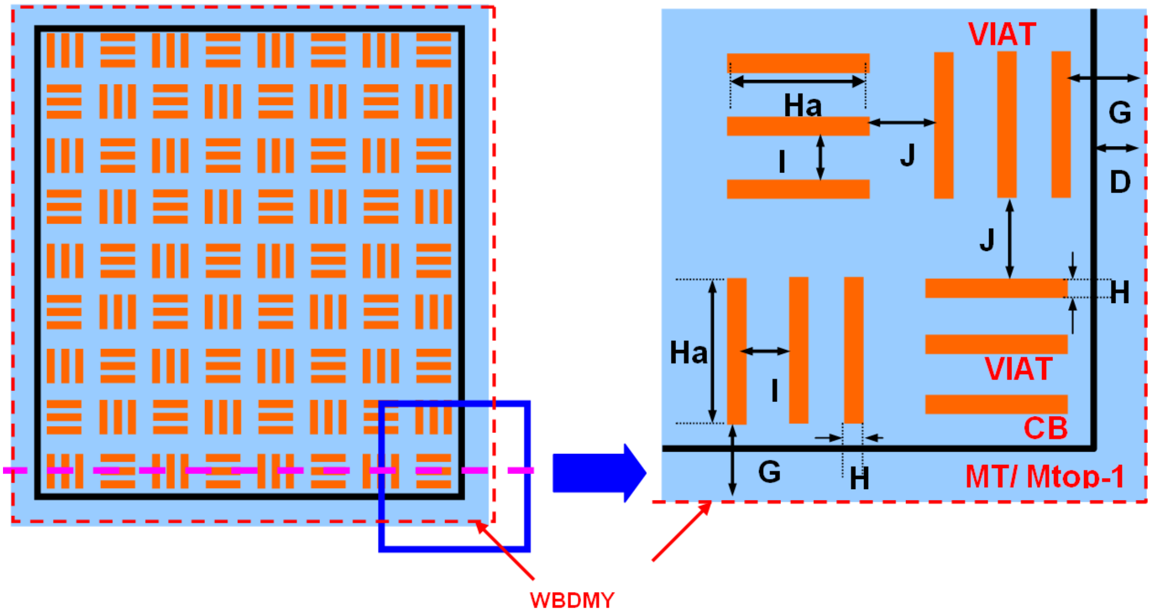
- **Triple Line PAD top via only size: 0.28um(H)X3.44um(Ha)**

Metal/Via/Passivation Opening Window (CB) Rules:

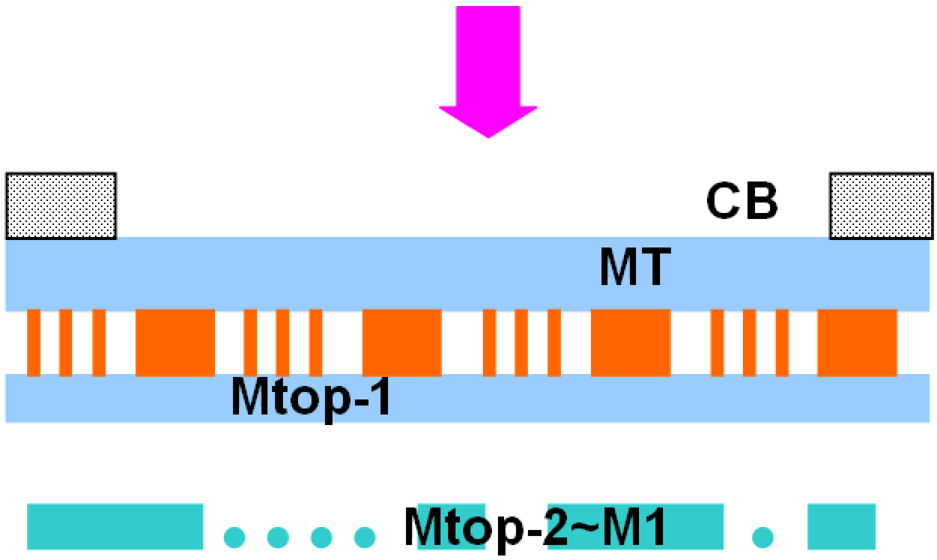
Rule No.	Description	Label		Rule
Common items (for Single In-Line/ Staggered/ Tri-Tier)				
CUPCB.EN.1	Enclosure by MT/Mtop-1, except CUP pad: M1~Mtop-2	D	≥	2
CUPCB.R.1	1. A specialized CUP pad structure must be applied to the top two metal layers with top via, Mtop/VIAtop/ Mtop-1. 2.ᵘ VIAtop pattern under CB region must be composed of the vertical and horizontal triple-line cells. Please refer to **Figure 3.2.4.3.1** and **Figure 3.2.4.3.2** for the correct top VIA drawing. 3.ᵘ WBDMY (183;0) must be drawn to cover pads for DRC checking. Please refer to the patterns of **Figure 3.2.4.3.1** and **Figure 3.2.4.3.2** for the correct WBDMY drawing.			
CUPVIAT.EN.1	Triple-line VIAtop enclosure by Mtop/ Mtop-1	G	≥	2
CUPVIAT.W.1	Width of line VIAtop	H	=	0.28
CUPVIAT.W.2	Length of line VIAtop	Ha	=	3.44
CUPVIAT.S.1	Space of two line VIAtop [in parallel direction] (within the same Triple-line VIA cell)	I	=	1.3
CUPVIAT.S.2	Space of two line VIAtop [in perpendicular direction] (between two neighboring Triple-line VIA cells)	J	=	2
CUPVIAT.DN.1	Ratio of total exposure line VIAtop area to CB area within CB region		≥	8%
CUPCB.R.3	Slots in Mtop and Mtop-1 pad region are not allowed, DRC check metal pad by the following definitions: (1) {Mx AND [CB sizing 3]}, or (2) {(Mtop/Mtop-1) AND [CB sizing 3]}			
TRL.R.1	It defined TrL as VIAtop in CUP, you also must use TrL deformed fully stacking PAD if non-CUP demand in whole chip.			

CUP Pad Structure with Triple Line cell VIA Schematic Diagram:

Top view:(Figure 3.2.4.3.1) (Figure 3.2.4.3.2)



Cross-Section:(Figure 3.2.4.3.3)



3.2.4.4 One Big Via (OBV) CUP Pad structure Rules

3.2.4.4.1 C025 OBV Design Rules

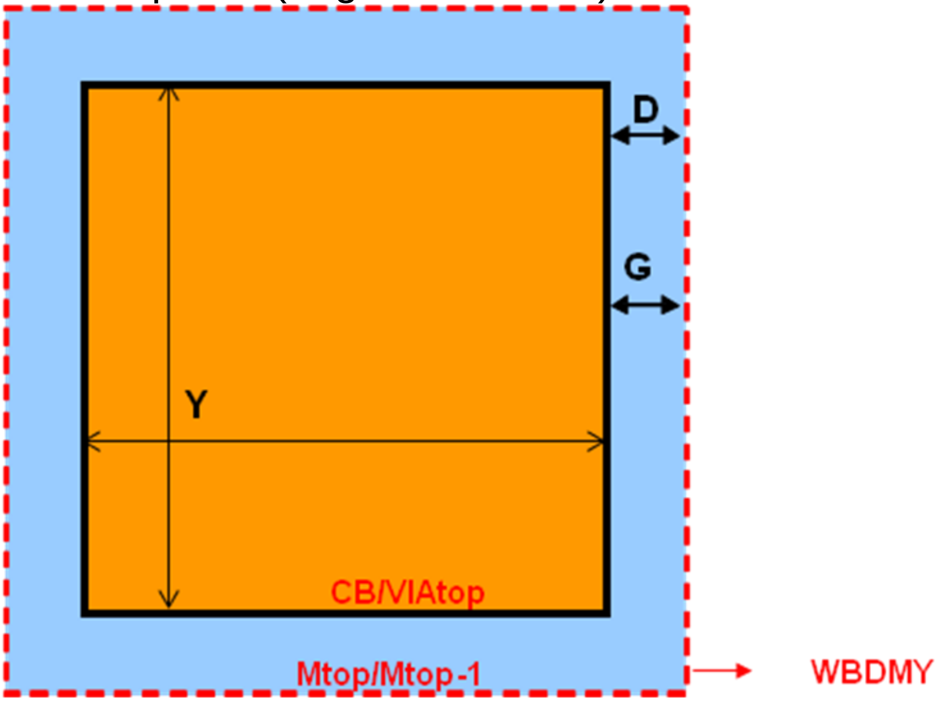
- **OBV pad is only used in single-in line PAD layout.**

Metal/Via/Passivation Opening Window (CB) Rules:

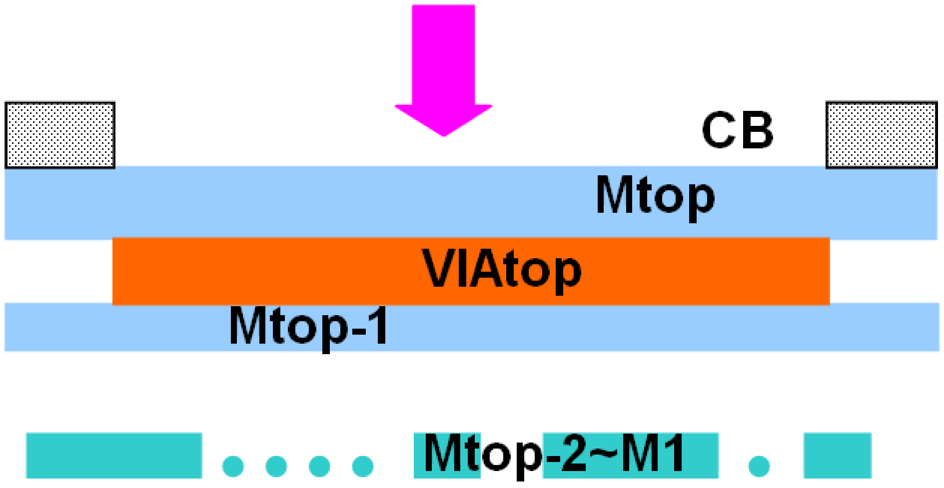
Rule No.	Description	Label		Rule
Common items (only Single In-Line)				
CUPCB.EN.2	CB enclosure by Mtop/Mtop-1.	D	≧	2
CUPCB.R.2	1. A specialized CUP pad structure must be applied to the top two metal layers with top via, Mtop/VIAtop/ Mtop-1. 2. VIAtop must be exactly identical to {(CB sizing 2) sizing -2} in CUP pad structure(Notch exclusion). Please refer to **Figure 3.2.4.4.1** and **Figure 3.2.4.4.2** for the correct top VIA drawing. 3.U WBDMY (183;0) must be drawn to cover PAD (PAD≡{CB sizing 2}) for DRC checking. Please refer to the patterns of **Figure 3.2.4.4.1** for the correct WBDMY drawing.			
CUPVIAT.W.3	The maximum width of top via in OBV CUP.	Y	≦	300
CUPVIAT.EN.1	VIAtop enclosure by Mtop/ Mtop-1	G	≧	2
CUPVIAT.DN.1	The maximum density of top via (all of top vias) across whole chip		<	10%
CUPCB.R.3	Slots in Mtop and Mtop-1 pad region are not allowed, DRC check metal pad by the following definitions: (1) {Mx AND [CB sizing 3]}, or (2) {(Mtop/Mtop-1) AND [CB sizing 3]}			
OBV.R.1	It defined OBV as VIAtop in CUP, you also must use OBV deformed fully stacking PAD if non-CUP demand in whole chip.			
OBV.R.2	It only allowed single in line PAD distribution in the whole chip.			

CUP Pad Structure with One Big VIA (OBV) Schematic Diagram:

OBV Top view(Figure 3.2.4.4.1)



OBV Cross-Section(Figure 3.2.4.4.2)



3.2.4.4.2 C018/C016/C0152/C015 OBV Design Rules

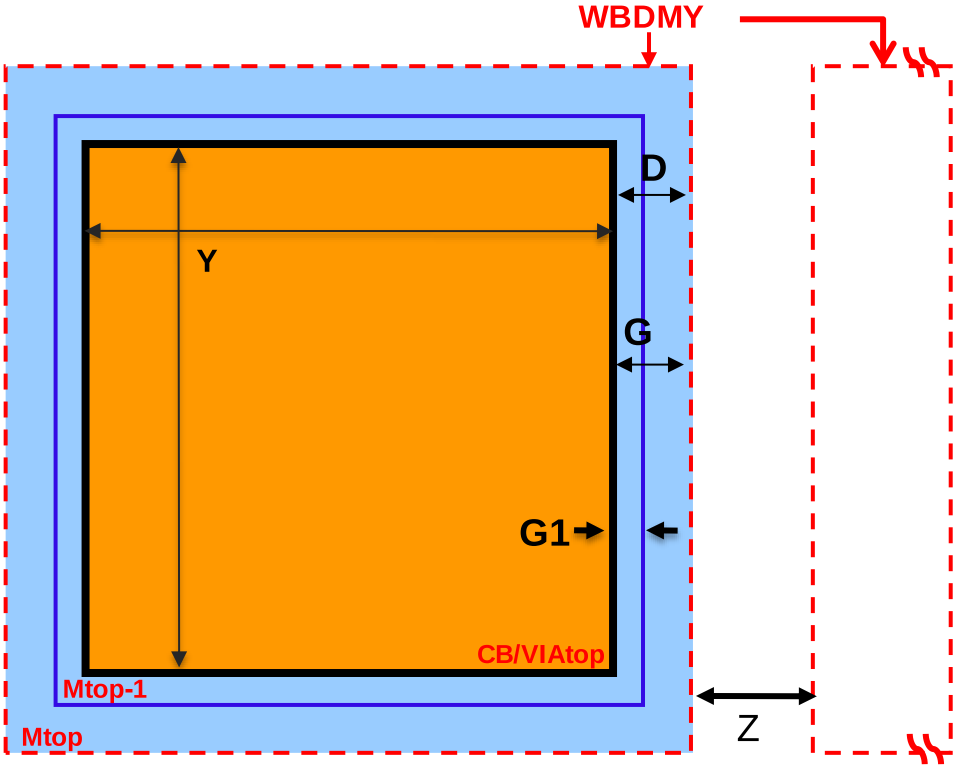
- **OBV pad is only used in single-in line PAD layout.**

Metal/Via/Passivation Opening Window (CB) Rules:

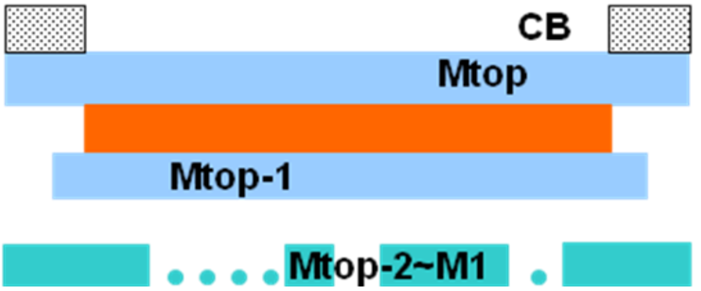
Rule No.	Description	Label		Rule
Common items (only Single In-Line)				
CUPCB.EN.2	CB enclosure by Mtop.	D	≥	2
CUPCB.R.2	1. A specialized CUP pad structure must be applied to the top two metal layers with top via, Mtop/VIAtop/ Mtop-1. 2. VIAtop must be exactly identical to {(CB sizing 2) sizing -2} in CUP pad structure (Notch exclusion). Please refer to **Figure 3.2.4.4.3** and **Figure 3.2.4.4.4** for the correct top VIA drawing. 3.ᵁ WBDMY (183;0) must be drawn to cover PAD region (PAD≡{CB sizing 2}) for DRC checking. Please refer to the patterns of **Figure 3.2.4.4.3** for the correct WBDMY drawing.			
WBDMY.S.1	The minimum space of WBDMY to WBDMY	Z	≧	3
CUPVIAT.W.4	The maximum width of top via in OBV CUP	Y	≦	150
CUPVIAT.S.3	The minimum space of one big via to one big via. One big via ={ Top via interact WBDMY}.	X	≥	7
CUPVIAT.EN.2	VIAtop enclosure by Mtop	G	≥	2
CUPVIAT.EN.3	VIAtop enclosure by Mtop-1 except butted metal interconnection. Metal width in {(CB SIZING 2) NOT (CB SIZING 0.2)} region must be less than the Mtop-1 pad dimension.	G1	=	0.2
CUPVIAT.DN.1	The maximum density of top via (all of top vias) across whole chip		<	10%
CUPCB.R.3	Slots in Mtop and Mtop-1 pad region are not allowed, DRC check metal pad by the following definitions: (1) {Mx AND [CB sizing 3]}, or (2) {(Mtop/Mtop-1) AND [CB sizing 3]}			
OBV.R.1	It defined OBV as VIAtop in CUP, you also must use OBV deformed fully stacking PAD if non-CUP demand in whole chip.			
OBV.R.2	It only allowed single in line PAD distribution in the whole chip.			

CUP Pad Structure with One Big VIA (OBV) Schematic Diagram:

OBV Top view (Figure 3.2.4.4.3)



OBV Cross-Section (Figure 3.2.4.4.4)



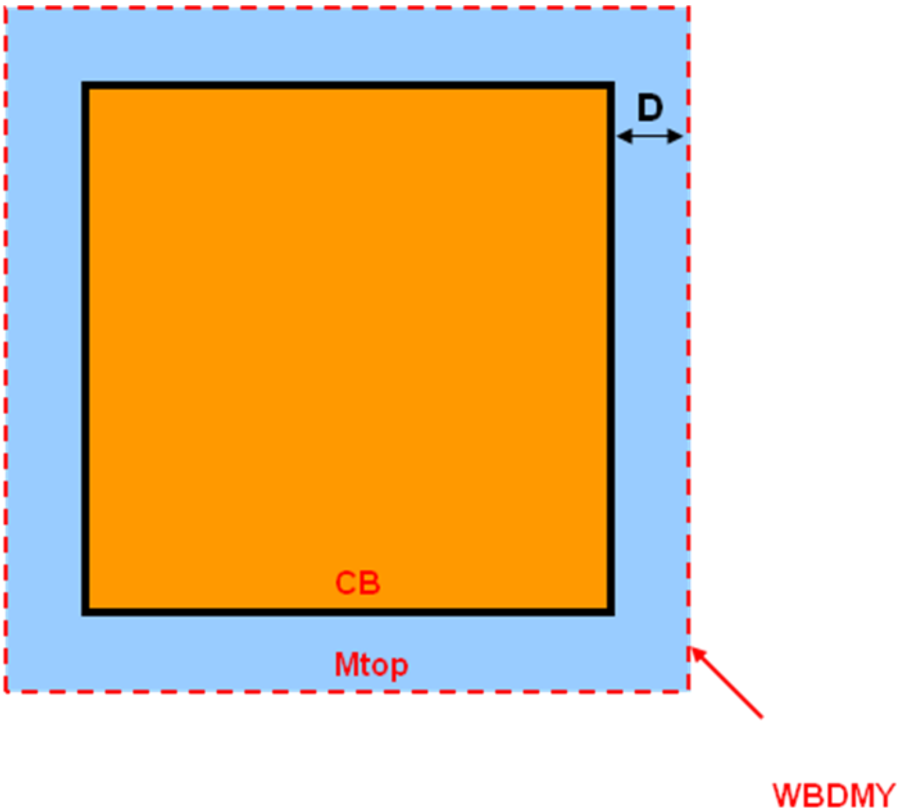
3.2.4.5 Single Metal (SM) CUP Pad structure Rules

- **Single Metal PAD top via size: N/A (No via pattern).**

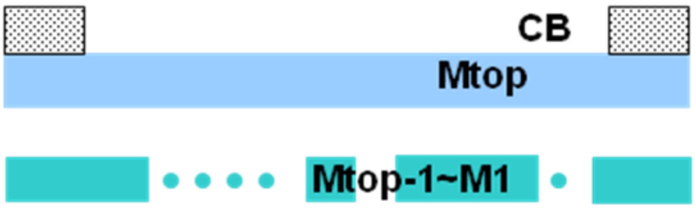
Rule No.	Description	Label		Rule
Common items (for Single In-Line/ Staggered/ Tri-Tier)				
CUPCB.EN.3	CB enclosure by Mtop.	D	≥	2
CUPCB.R.4	1. A specialized CUP pad structure must be applied to the top metal layers without top via. 2. WBDMY (183;0) must be drawn to cover pads for DRC checking. Please refer to the patterns of **Figure 3.2.4.4.7** for the correct WBDMY drawing.			
CUPCB.R.3.1	Slots in Mtop pad region are not allowed, DRC check metal pad by the following definitions: (1) {Mx AND [CB sizing 3]}, or (2) {(Mtop/Mtop-1) AND [CB sizing 3]}			
SM.R.1	It defined SM as no VIAtop in CUP, be not used non-CUP demand in whole chip, when 1P2M scheme.			

CUP Pad Structure with Single Metal (SM) Schematic Diagram:

SM Top view:(Figure 3.2.4.4.7)



SM Cross-Section:(Figure 3.2.4.4.8)

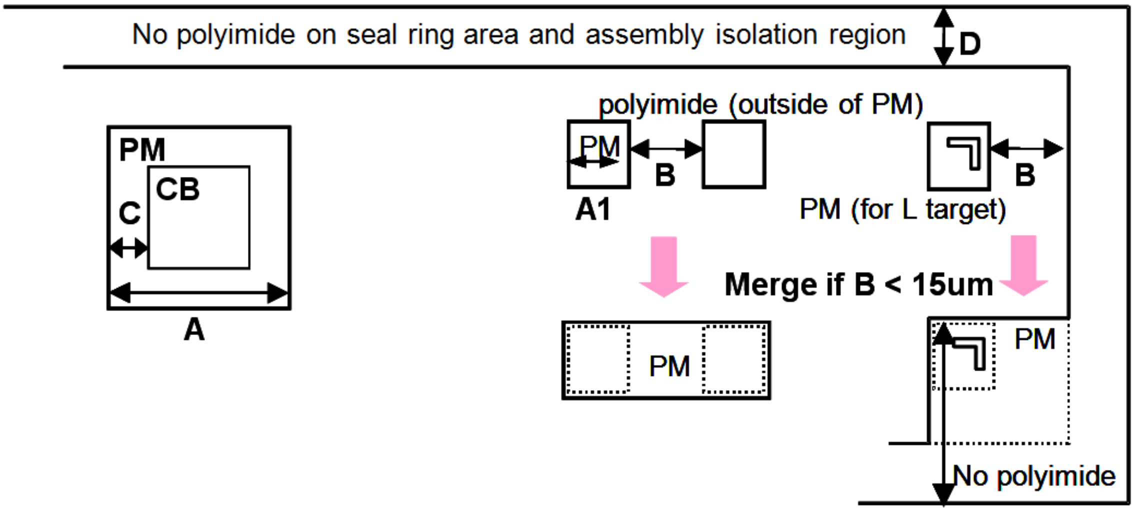


3.2.5 Polyimide (PM) Rules for Wire Bond

- Polyimide is required for large chip sizes (> 60 mm^2 or > 100 mm^2 for low stress material packages, for example, BGA).
- **Polyimide** is an option for C025~C018 and its half node technologies.
- The rules are based on TSMC process capability. If you want to implement polyimide layer at the assembly house, please consult with their rules.
- By default, the PM mask is generated by logic operation from CB with the reverse tone of CB mask.
- The following rules must be followed if customers draw the Polyimide layout by themselves.

Rule No.	Description	Label		Rule
Common items (for Single In-Line/Staggered/ Tri-Tier)				
PM.W.1	Width (Interact with CB region)	A	≥	86
PM.W.2	Width (Not interact with CB region)	A1	≥	30
PM.S.1	Space of two PM, or space of PM to chip edge. Merge if space is less than 15 µm.	B	≥	15
PM.EN.1	Enclosure of CB	C	≥	5
PM.R.1 u	Chip size that can do without polyimide for non-low stress material package.		≤	60 mm^2
PM.R.2 u	Chip size that can do without polyimide for low stress material package (for example, BGA)		≤	100 mm^2
PM.R.3	Polyimide is prohibited over the seal ring area and assembly isolation region, LMARK and FW	D		

PM:



3.2.6 Wire Bond Non-shrinkable Rules for the Half Node Technologies

- The layout dimensions of the following items need to be sized up before 90% shrinkage. For example, C018 → C016

Rule No	Description	Rule			
			Pitch ≥ 50μm‡ Single in-line	Pitch ≥ 55μm‡ Single in-line	Pitch ≥ 60μm Stagger
#CB.W.1	Width	≥	47.3	52.8	58
#CB.W.2	Length of (the edge perpendicular to nearby chip edge)	≥	88	66†	66†
#CB.S.1	Space	≥	7†	7†	7.7
#CB.P.1u	Pitch of (4 pitches at each chip corner)	≥	88	88	110

Table notes:

- **† These rules are shrinkable.**
- **‡ The pitch is real pad pitch after 90% shrinkage.**

4 Layout Rules for Flip Chip

Need to perform package qualification for change in wafer technology, wafer passivation scheme, die size, package type, package size and package material.

4.1 Recommendations for Flip Chip

4.1.1 General Layout Recommendations

1. **Ground-up (Area-array I/O):** *Ground-up* refers to the interconnect layout style of a flip chip design in which the I/O cell and solder bump pads are distributed within the chip core (not placed in a ring outside the core) to take advantage of a minimum routing distance and chip size. Therefore, there is no redistribution layer required for a flip chip product with ground-up design only.

   The ground-up layout style is also known as an *area-array I/O* because the I/O circuits must be arranged in an array to minimize the distance between the I/O and the metal pads. The following information should be noted regarding the ground-up design at TSMC:

   a. The wafer process flow of ground-up (area-array I/O) products at TSMC's fab is identical to that of the wire bond products.

   b. Because the I/O circuits are no longer restricted to the periphery of the chip, there are things to be aware of to ensure good ESD and latch-up performance. For details, please refer to “I/O ESD protection circuit design and layout guideline” in the appropriate logic design rule manual.

2. **Die size and bump pad pitch selection:** Please take assembly process capability and packaging type into consideration. In the early design stage, it is very important to consider the bump pitch rules in your chip layout in conjunction with the substrate layout at the same time, especially for the small bump pitch design below 160um pitch. Please consult with your assembly house for the substrate information and process capability first. TSMC recommends following Table 4.1.1 to determine the proper bump pitch, UBM size and bump materials.

Table 4.1.1 Die size to bump pitch, UBM width and bump material selection

Chip Area (mm^2)	**Area≦100**	**100<Area≦225**	**225<Area≦400**	**Area>400**
Min. Bump Pitch (um)	**150**	**160**	**180**	**200**
Min. UBM (um)	**80**	**85**	**90**	**100**
Bump Composition*	**EU, HL**	**EU, HL**	**EU**	**EU**

* EU: eutectic, HL: high lead

*** Warning:** For design with a bump pitch below 160um, please consult your assembly house in advance. Make sure that your assembly house is able to provide such substrates and the associated service for your smaller bump pitch design.

3. Required bumping and testing flow:

a. TSMC's standard backend process for flip chip products is:

PM or passivation opening (fuse window [if applicable] etch) ⇨ WAT ⇨(Fab out)⇨ Solder bumping ⇨ Die sort (CP1) ⇨ Laser repair [if applicable]⇨ Die sort (CP2) [if applicable] ⇨ Flip chip assembly ⇨ Final test (FT)

As a result, "die sort before bumping" is prohibited. Please contact TSMC for further technical support if needed.

b. TSMC only allows HL/EU bump products at 150℃ test. If customers need higher baking temperature, please consult with TSMC.

Note: TSMC products' qual pass HTS/150℃ 1000hrs reliability test.

4. Mask making: The below masks can't be generated by logical operation for flip chip design. Customers should provide the mask drawings in the tape-out GDSII file.

- CB/CBD (mask107)
- PM (mask 009)
- UBM (mask 020)

5. Bumping and UBM (Under Bump Metallurgy) masks: For bumping in TSMC, the UBM mask is needed for tape-out. For bumping in third-party bump houses, a UBM mask is not needed for tape-out to TSMC. Please consult with bump houses for their detailed UBM rules.

6. Polyimide (PM):

- **Polyimide** is not a standard offering.
- If a customer doesn't use TSMC bump service, the PM layer should be implemented at your bump house, not at TSMC. This is to prevent the unpredictable compatibility problem that might happen between the PM and the bumping process.

7. Backside grinding: Wafer backside grinding is not recommended before bumping because of concerns about wafer breakage incurred in the bumping process. Therefore, backside grinding is allowed to a minimum 482.6μm (19 mils) thickness for 8" wafers.

8. Alpha particle sensitive area: Bumps on the top of the alpha particle sensitive circuit, like SRAM, are not recommended, except when ultra low alpha particle materials (Solder bump, under-fill, pre-solder bump…) are used. TSMC bump line uses ultra low alpha particle material. If the bumping is not done at TSMC, please consult with your bumping house about the alpha particle emission rate data and make sure your assembly house uses ultra low alpha particle materials. Please refer to the rules of UBM.R.4g and UBM.S.4® in sec.4.4.

9. Dummy bumps: Bumps that are farther away from the center of a die are generally more susceptible to higher mechanical stress. Dummy bumps that are distributed over the chip and that follow regular design rules are recommended for improving mechanical and thermal performance.

10. Layout under bump: Device placement and routing under bump pads is allowed.

11. Orientation identifier: For a chip with a perfectly symmetrical bump array, an orientation or a "marker" bump is recommended to create asymmetry and to avoid pick-and-place issues during flip chip assembly processes.

12. Polyimide (PM)/CBD window opening:

Whenever possible, use a larger PM, CBD opening.

4.1.2 Recommendations for Redistribution Metal

1. **Redistribution layer:** *Redistribution* refers to the rearrangement from the traditional peripheral **wire bond pad to the solder bump pad array** by an additional interconnection metal, termed a redistribution layer, which is on the top of the original top metal (Mtop). A redistribution layer is used to convert a product from a wire bond to a flip chip package.

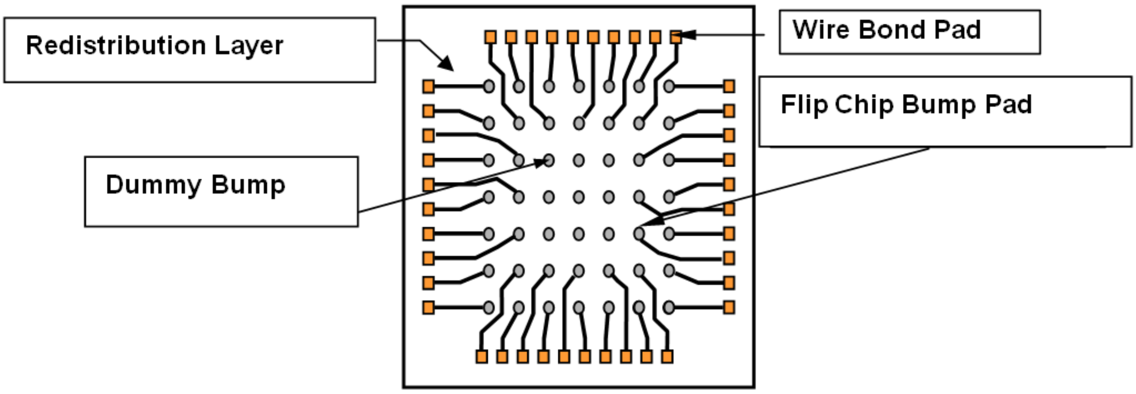


Figure 4.1.2 Schematic diagram of bump pad redistribution

2. **Interconnection for a solder bump of a flip chip package:** Please refer to section 4.2. Two kinds of interconnection are offered for those who want to design the interconnection for a solder bump of a flip chip package.
   a. **RDL (redistribution before passivation):**

      This interconnection style uses additional via (VIAD) and metal (MD) layers above the original top metal (Mtop). The VIAD and MD design rules and processes (film thickness and mask grade) are identical to the original ones of VIAtop and Mtop. Please refer to section 4.3.6.

   b. **PPI (redistribution after passivation1 and passivation2):**

      This kind of process uses CB as a via hole to connect Mtop and PPI. Please refer to the rules in the sec. 4.2.7.

      - Additional metal layer after passivation. (mask name: PPI, mask_ID 169). Please refer to section 4.2.7.2.

3. **I/O ESD protection recommendations:**

   a. Recommended minimum bus line width is Al: 20µm. The bus line width definition is the total width of all metal lines (no matter MD, PPI) connecting with the same bump pad.

   b. Recommended maximum BUS line resistance from V_{dd} pad or V_{ss} pad to any I/O pad is 3Ω.

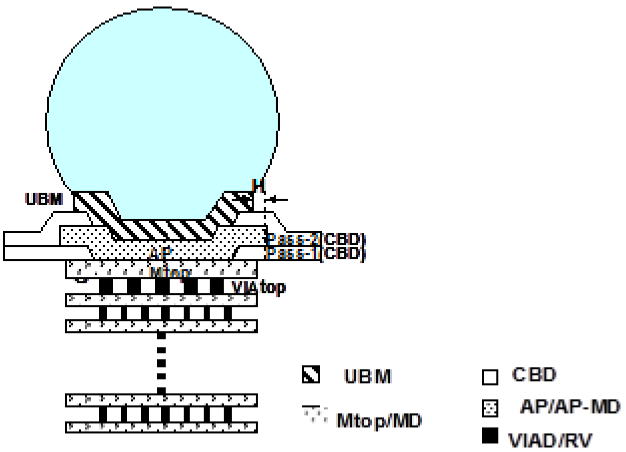
   c. Recommended minimum counts of VIAD connecting to bump pad:

   C025~C015: 200 each

4. **Please refer to the "I/O ESD Protection Guideline" section in the corresponding design rule manual.**

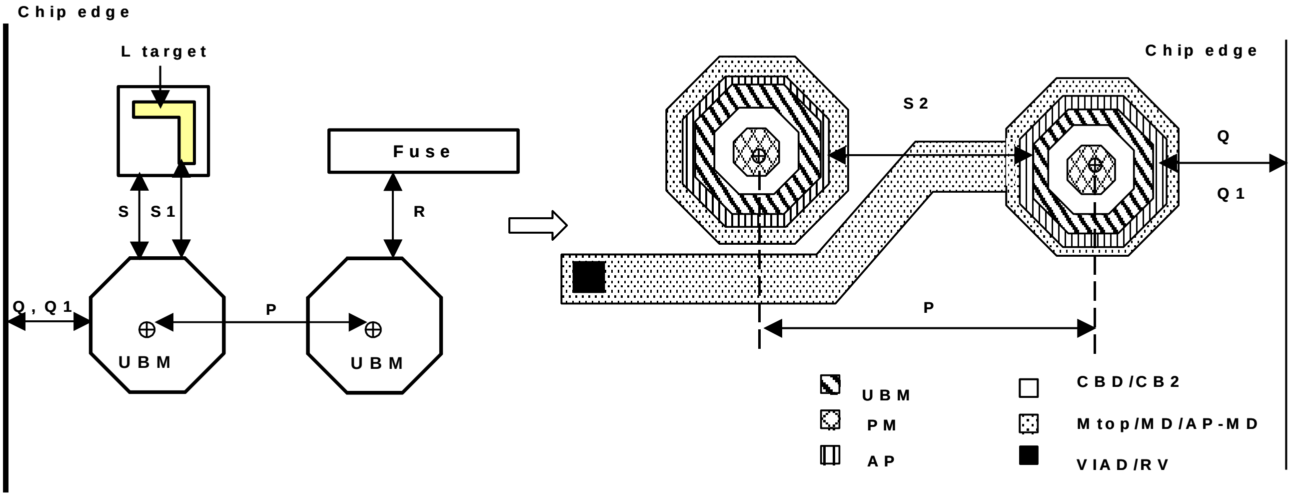
4.2 Under Bump Metallurgy (UBM) Rules

Rule No.	Description	Label		Rule
UBM.P.1	Pitch	P	≥	Table 4.3.3.1.b
UBM.S.1	Space to metal fuse protection ring	R	≥	50
UBM.S.2	Space to L target protection ring	S	≥	60
UBM.S.4®	Space of two UBMs (Maximum) It is recommended not to have isolated bumps, or avoid unnecessarily spare regions.	S2	≤	400
UBM.EN.1	Enclosure by chip edge	Q	≥	80
UBM.EN.1®	Recommended enclosure by chip edge (Maximum)	Q1	≤	300
UBM.EN.2	Enclosure by Mtop/ MD/ PPI	H	≥	2
UBM.A.1®u	Area (without UBM in the chip center, for test inking requirement. Not necessary for inkless sorting)		≥	750x750
UBM.DN.1	Density across full chip Dummy UBM is required if the UBM density is less than spec. The dummy pattern should be as uniform as possible over the chip.		≥	10%
UBM.DN.3	Minimum area ratio, calculated by {UBM sizing +200um} divided by the whole chip area. Only one single polygon is allowed after the {UBM sizing +200um} operation. Please refer to Figure 4.2.2.		≥	80%
UBM.R.1	Shape of UBM/CBD/ PM for bump: equilateral octagon is minimum requirement, and circle shape is not recommended. Only single size bump/UBM is allowed. DRC only check the minimum vertexes of the equilateral polygon: 8.			
UBM.R.2 u	Dummy UBM that follow rules are required to improve mechanical strength and thermal dissipation.			



<table>
<tr><th>Rule No.</th><th>Description</th><th>Label</th><th></th><th>Rule</th></tr>
<tr><td>UBM.R.3</td><td>Bump ball (includes active bump and dummy bump) structure must include the layers (at least) in the below table. UBM pattern not for bump (for example, seal ring, part ID, fuse window, L target window, text mark or logo) is forbidden.
<table>
<tr><td>Technology</td><td>Mtop</td><td>MD</td><td>CBD</td><td></td><td></td><td>PM*</td><td>UBM</td></tr>
<tr><td>C015~C025</td><td>√</td><td></td><td>√</td><td></td><td></td><td>√</td><td>√</td></tr>
<tr><td>C015~C025
With MD</td><td></td><td>√</td><td>√</td><td></td><td></td><td>√</td><td>√</td></tr>
</table>
* If you use polyimide in your design, the bump ball structure must include the PM layer as well.</td><td></td><td></td><td></td></tr>
<tr><td>UBM.R.5</td><td>A minimum number of chip corner dummy bumps is required according to the chip size; these dummy bumps must be the ones that are nearest to chip corner among all bumps. Please refer to Figure 4.2.1 and Table 4.2.1.
DRC checks the chip corner dummy bump by the following conditions:
1. The DRC deck defines a "UBM bounding box" for the whole chip by forming a smallest rectangle that contains all UBM bump pads. (Figure 4.2.1.)
2. At the corner of the “UBM bounding box”, a “corner bump check zone” is then determined according to the chip size. (Table 4.2.1 defines the leg dimension of the corner bump zone.)
3. Check how many “chip corner bumps” interact with the “corner bump check zone”
4. Check whether the "chip corner bumps" are dummy bumps.
5. Dummy bump definition:
Without RDL (ground-up): Top metal of bump pad structure does not connect to any top VIA.
With MD RDL: MD of bump pad structure does not connect to any top VIAD.
With PPI RDL: PPI of bump pad structure does not connect to any RV or CBD.</td><td></td><td></td><td>Table 4.2.1</td></tr>
<tr><td rowspan="2">UBM.R.6® u</td><td rowspan="2">Recommended size ratio of UBM/Pre-Solder Bump SRO (=C/N) and BGA SRO/Board Pad (=T/O). Ratio=1.05 is preferred.
SRO: Solder Resist Opening</td><td rowspan="2"></td><td>≥</td><td>0.95</td></tr>
<tr><td>≤</td><td>1.05</td></tr>
<tr><th>Guideline</th><th colspan="4">Description</th></tr>
<tr><td>UBM.R.4g u</td><td>It is recommended not to put any bump on the top of SRAM, analog, sensitive circuits, and the matching pairs.
The circuits should be located at a minimum distance of 60 µm from the bump pad's PM or CBD edge.
It is also recommended to consider UBM.S.4® at the same time.
If bump over SRAM, analog, or sensitive circuit areas is needed, it is recommended to use the ultra-low alpha particle materials in the bump and assembly processes (solder bump, under-fill, pre-solder bump…) to avoid a high Soft Error Rate (SER).
TSMC uses ultra-low alpha particle materials in the solder bump process.
If a customer could not meet UBM.S.4® and UBM.R.4g at the same time, you can consult TSMC for the layout suggestions.</td><td></td><td></td><td></td></tr>
<tr><td>UBM.R.5g u</td><td>It is recommended not to place the IO bump pads in the 2nd and 3rd row in the bump array corner, but put redundant Vss, Vdd, or dummy bump pads.</td><td></td><td></td><td></td></tr>
<tr><td>UBM.R.7®</td><td>Recommended UBM width selection according to chip size.</td><td></td><td></td><td>Table 4.3.3.2</td></tr>
</table>

UBM:
Top View



Schematic diagram for UBM.R.6®

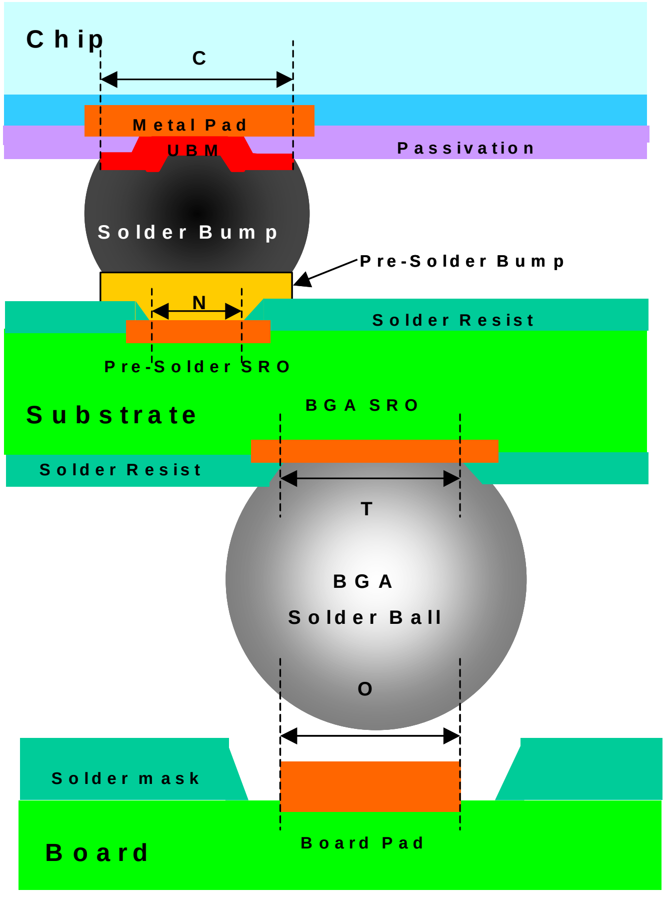


Figure 4.2.1 Schematic diagram for UBM.R.5.

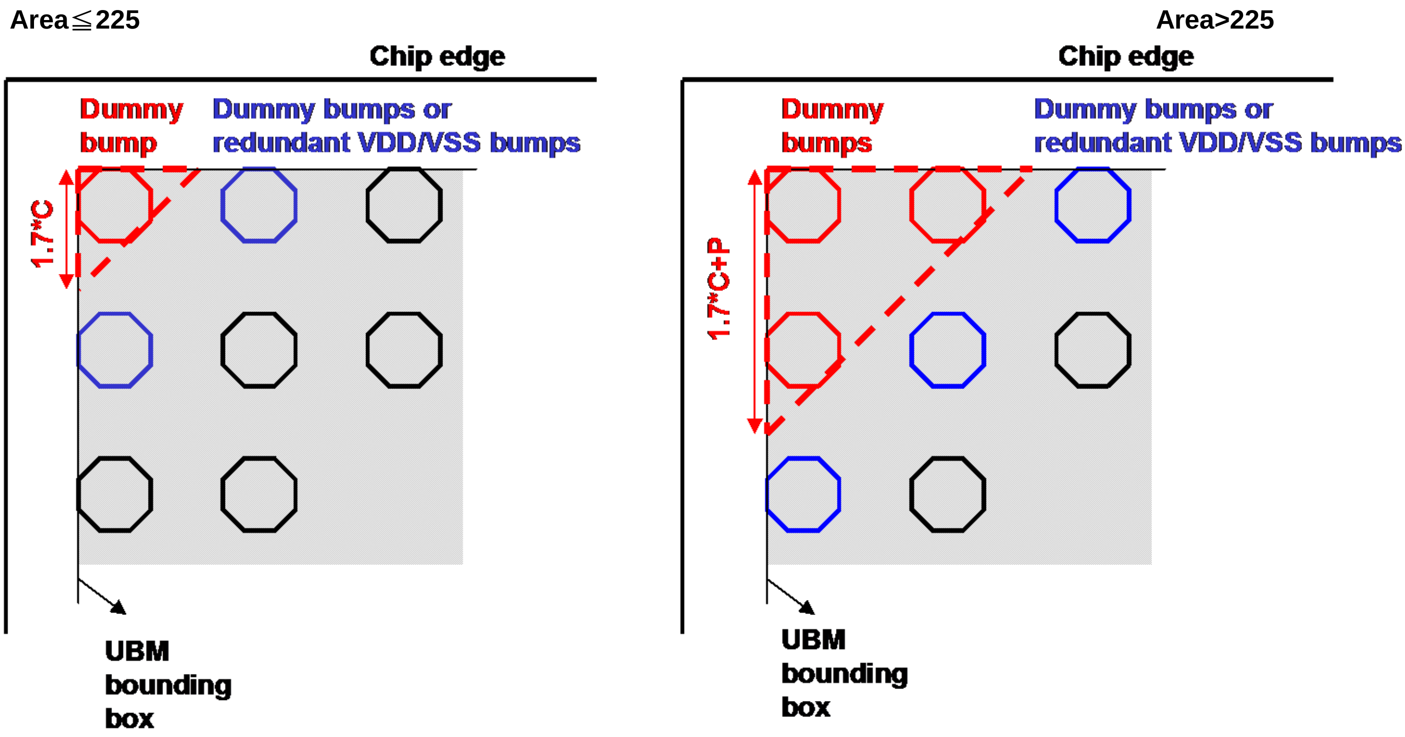


Table 4.2.1 Area of active bump exclusion zone and dummy bump counts.
(1) Main node (Layout = On-Si)

Design Dimension = On Silicon Dimension (μm)	Chip size (mm^2)	Area≦225	Area > 225
	Leg dimension of corner bump check zone	1.7*C	1.7*C+P
	Minimum dummy bump count	1	3

(2) Half node (111% blow up from main node (half node On-Si=100%; Layout=111%)

Design Dimension (μm)	Chip size (mm^2)	Area≦278	Area > 278
On Silicon Dimension (μm)	Chip size (mm^2)	Area≦225	Area > 225
	Leg dimension of corner bump check zone	1.7*C	1.7*C+P
	Minimum dummy bump count	1	3

C: UBM width; P: min. UBM pitch according to the chip size

Schematic diagram for UBM.R.5g

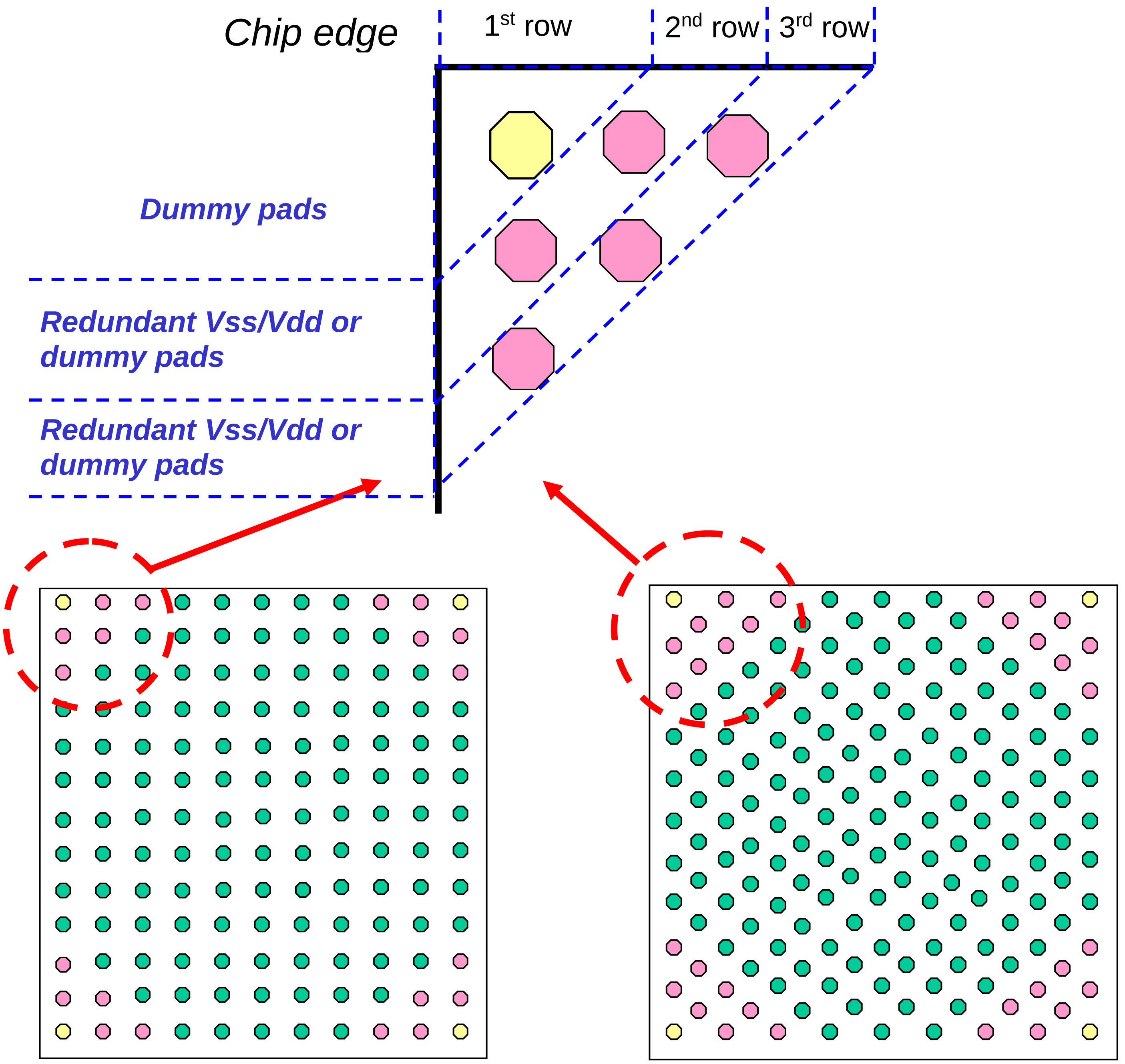
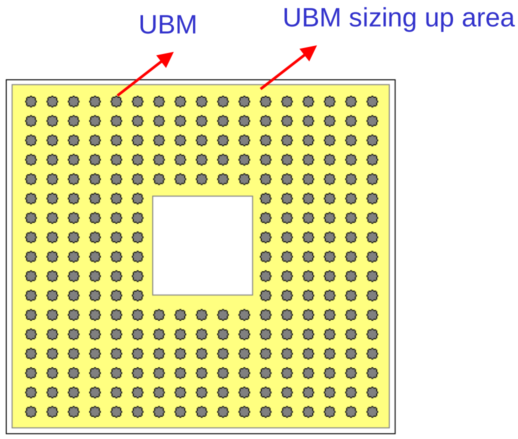
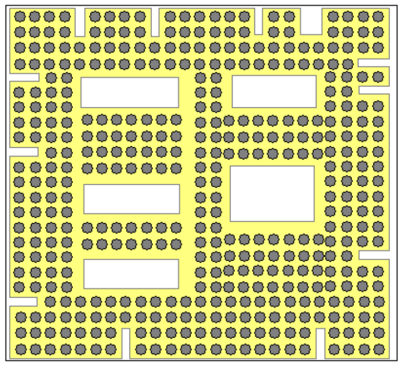


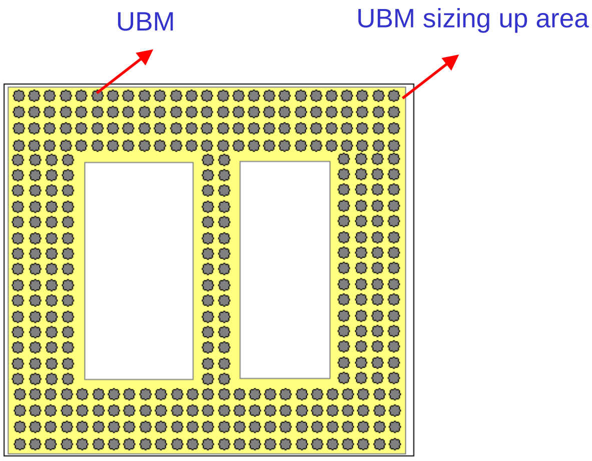
Figure 4.2.2 Schematic diagram for UBM.DN.3



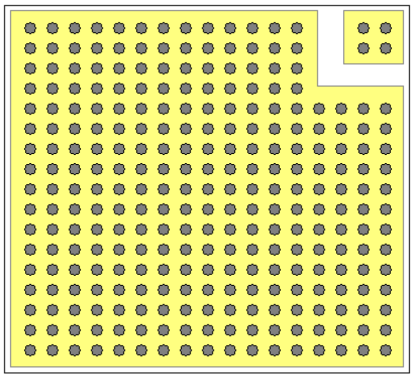
OK
(UBM sizing up area density ≥ 80%)



OK
(UBM sizing up area density ≥ 80%)



Bad
(UBM sizing up area density < 80%)



Bad
(Two separate polygons after sizing up)

4.3 Flip Chip Rules (C025~C015)

4.3.1 Mask Information:

Table 4.3.1.1 Mask Information and CAD Layers (C025/C022/C018/C015)

Sequence	Mask Name	Mask ID	Digitized Area (Dark or Clear)	CAD Layer** (C018/C022 /C025 only)	Reference Layer in Logical Operation	Type			Description
						FC1	FC2	FC3	
1	VIAD	‡	C	‡	‡	**0**	**1**	**0**	Via hole between MD and Mtop
2	MD	‡	D	168	‡	**0**	**1**	**0**	Redistribution metal for flip chip
3	CBD	107	C	169	-	**1**	**1**	**1**	Passivation pad opening for flip chip
4	PPI	169	D	189	-	**0**	**0**	**1**	Post passivation interconnection for flip chip
5	PM	009	D	89	-	*****	*****	**1**	Polyimide window for flip chip
6	UBM	020	D	170	-	**1**	**1**	**1**	Under bump metallurgy

FC1	**Flip chip without RDL (MD)**
FC2	**Flip chip with RDL (MD)**
FC3	**Flip chip with PPI**

The definition of legends (0, 1, and *):

0 : Does not use the mask
1 : Must use the mask
*** : Optional mask**

‡: Follow the mask and CAD layer of the metal/VIA layer above the Mtop/VIAtop.

Warning: Some of the CAD layer numbers for C015 and other processes (C018/C022/C025) are different. The CAD layer provided in Table 4.3.1.1 is only for C018/C022/C025. Please align the correct CAD layer for each technology with the associated technology file.

**** CAD Layers Table (C025/C022/C018/C015):**

Mask Name	Mask ID	Digitized Area (Dark or Clear)	CAD Layer (C015 only)	CAD Layer (C018 only)	CAD Layer (C022 only)	CAD Layer (C025 only)
MD		D	168	168	168	168
CBD	107	C	169 (CBD)	169 (CBD)	169 (CBD)	169 (CBD)
PPI	169	D	42	189	189	189
PM	009	D	5	89	89	89
UBM	020	D	170	170	170	170

4.3.2 Reference Document

Content	Reference Documentation
Qualification report	• S-QCS-04-03-005 *POST CP WAFER QUALITY ASSURANCE SPECIFICATION (BUMP & NON-BUMP WAFER)* • T-015-LO-PF-011 *TSMC 0.15UM LOGIC 1P7M SALICIDE 1.2V/3.3V &1.5V/3.3V CU-FSG REDISTRIBUTION LAYER (RDL) PROCESS FLOW* • T-025-LO-PF-011 *TSMC 0.25UM LOGIC RDL (REDISITRIBUTION LAYER) 1P6M BRIEF PROCESS FLOW*
BUMP EM	• Q-RAS-02-02-083 *BUMP EM IMAX RULE (FOR TSMC'S BUMP LINE PROCESS ONLY)*

4.3.3 Bump Pad Structure Rules(C025~C015)

- The following bump pad rules are used for the TSMC bumping process. If you don't use TSMC bump service, please consult your bumping house regarding these rules.
- Ground-up and MD (redistribution before passivation) can be used on the same chip.
- There are five kinds of combinations for the bump and redistribution.

Items	Ground-up	Redistribution	
		Before passivation (RDL)	After passivation (PPI)
with PM	**V**	**V**	**V**
without PM	**V**	**V**	**x**

Bump Rules for Al Process With or Without Polyimide

Rule No.	Description	Label		Rule
BP.W.1	Width of CBD under UBM area	A	≥	40
BP.W.2	Width of PM under UBM area (Polyimide process only)	B	≥	30
BP.W.3	Width of UBM (CBD width and PM width must conform to Table 4.3.3.1)	C	=	Table 4.3.3.1
BP.EN.1	PM enclosure by UBM (Polyimide process only) PM not interact UBM is not allowed (except sealring, fuse, L-target).	D	≥	15
BP.EN.2	CBD enclosure by UBM (Without polyimide process only) CBD not interact UBM is not allowed (except sealring, fuse, L-target).	E	≥	10
BP.EN.3	CBD enclosure by Mtop/MD	G	≥	12
BP.EN.4	PM enclosure by CBD (Polyimide process only)	I	≥	10
UBM.EN.2	UBM enclosure by Mtop/ MD/ PPI	H	≥	2
BP.R.2®	Recommended bump pitch selection according to chip size.			Table 4.3.3.2

Table 4.3.3.1.a EU/HL Bumping Pad Dimensions (Unit in µm) for C016, Half node (111% blow up from main node (C016 On-Si = 100%; Layout (Design) = 111%))

Design Dimension (um)	P	Bump pitch	165-194*	194-222			222-250					≥ 250				
	C	UBM width	88**	88**	94.4	100	100	111		120		111		120		
	A	CBD width	44-66.6	44-66.6	44-72.2	44-77.7	44-77.7	44-88		44-97.7		44-88		44-97.7		
	B	PM width (with polyimide)	33-55	33-55	33-60.5	33-66	33-66	33-77		33-85.8		33-77		33-85.8		
On Silicon Dimension (um)	P	Bump pitch	150-175*	175-200			200-225					≥ 225				
	C	UBM width	80**	80**	85	90	90	100		108		100		108		
	A	CBD width	40-60	40-60	40-65	40-70	40-70	40-80		40-88		40-80		40-88		
	B	PM width (with polyimide)	30-50	30-50	30-55	30-60	30-60	30-70		30-78		30-70		30-78		
	Z	Bump height	80	80	85	90	90	90	100	90	100	90	100	90	100	110
	X	Bump diameter†	102	102	110	112	112	118	128	116	132	118	128	116	132	139

Table 4.3.3.1.b Al Process Bumping Pad Dimensions (µm)

P	Bump pitch	150-175*	175-200			200-225					≥ 225			
C	UBM width	80**	80**	85	90	90	100		108		100		108		
A	CBD width	40-60	40-60	40-65	40-70	40-70	40-80		40-88		40-80		40-88		
B	PM width (with polyimide)	30-50	30-50	30-55	30-60	30-60	30-70		30-78		30-70		30-78		
Z	Bump height	80	80	85	90	90	90	100	90	100	90	100	90	100	110
X	Bump diameter†	102	102	110	112	112	118	128	116	132	118	128	116	132	139

†Bump diameter depends on bump height for reference use

Warning: For design with a bump pitch below 160um, please consult your assembly house in advance. Make sure that your assembly house is able to provide such substrates and the associated service for your smaller bump pitch design.

** **Warning:** Design with a UBM width of 80um is still under evaluation at TSMC. Please consult TSMC first if you want to use such an 80 um UBM width.

Table 4.3.3.2.a Die size to bump pitch, UBM width and bump material selection for C016, half node (111% blow up from main node (C016 On-Si = 100%; Layout (Design) = 111%))

Design Dimension (um)	Chip Area (mm²)	Area≦123	123<Area≦278	278<Area≦494	Area>494
	Min. Bump Pitch (um)	165	178	200	222
	Min. UBM (um)	88	94.4	100	111
On Silicon Dimension (um)	Chip Area (mm²)	Area≦100	100<Area≦225	225<Area≦400	Area>400
	Min. Bump Pitch (um)	150	160	180	200
	Min. UBM (um)	80	85	90	100
	Bump Composition*	EU, HL	EU, HL	EU	EU

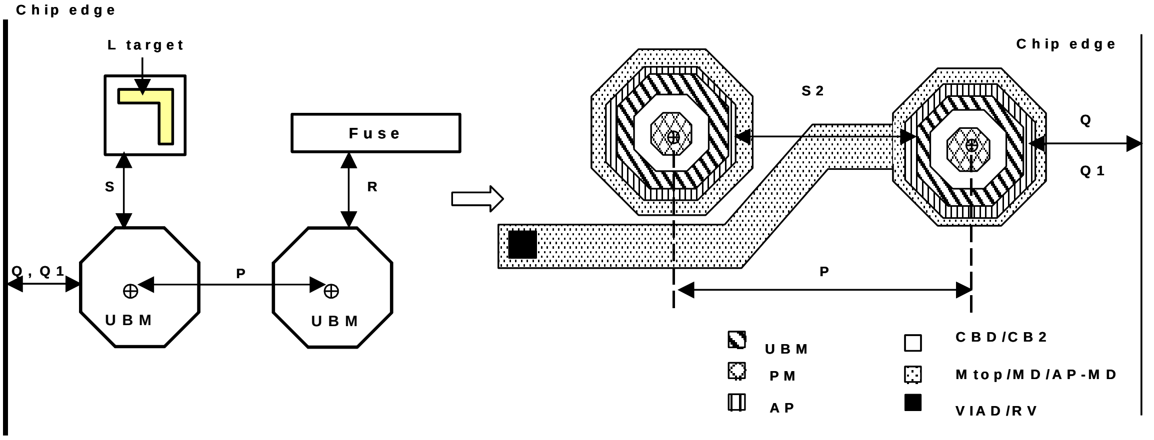
*** EU: eutectic, HL: high lead**

Table 4.3.3.2.b Die size to bump pitch, UBM width and bump material selection

Chip Area (mm²)	Area≦100	100<Area≦225	225<Area≦400	Area>400
Min. Bump Pitch (um)	150	160	180	200
Min. UBM (um)	80	85	90	100
Bump Composition*	EU, HL	EU, HL	EU	EU

* EU: eutectic, HL: high lead

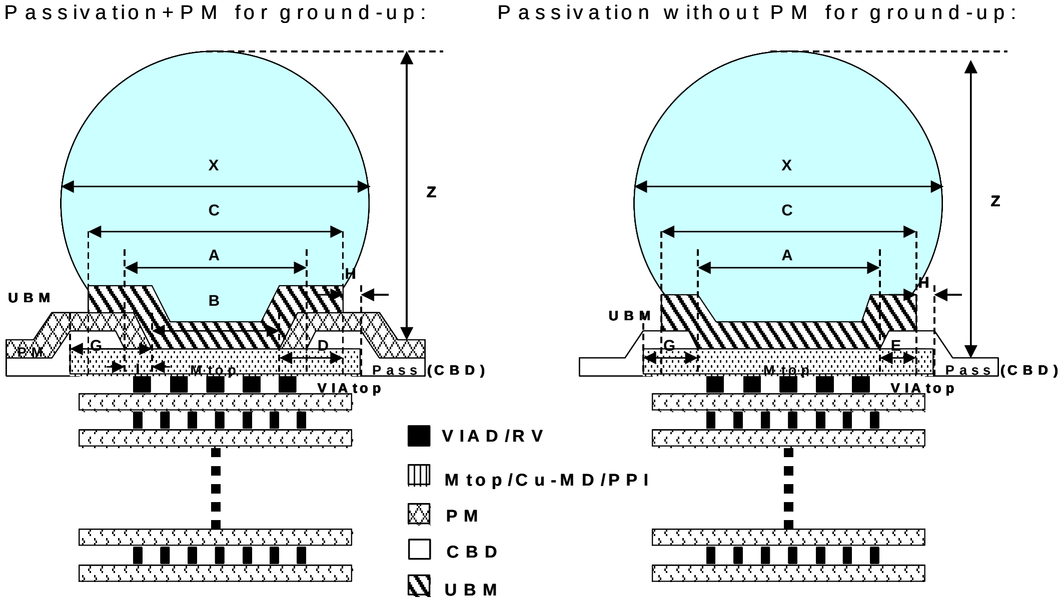
Bump Structure Schematic Diagram:



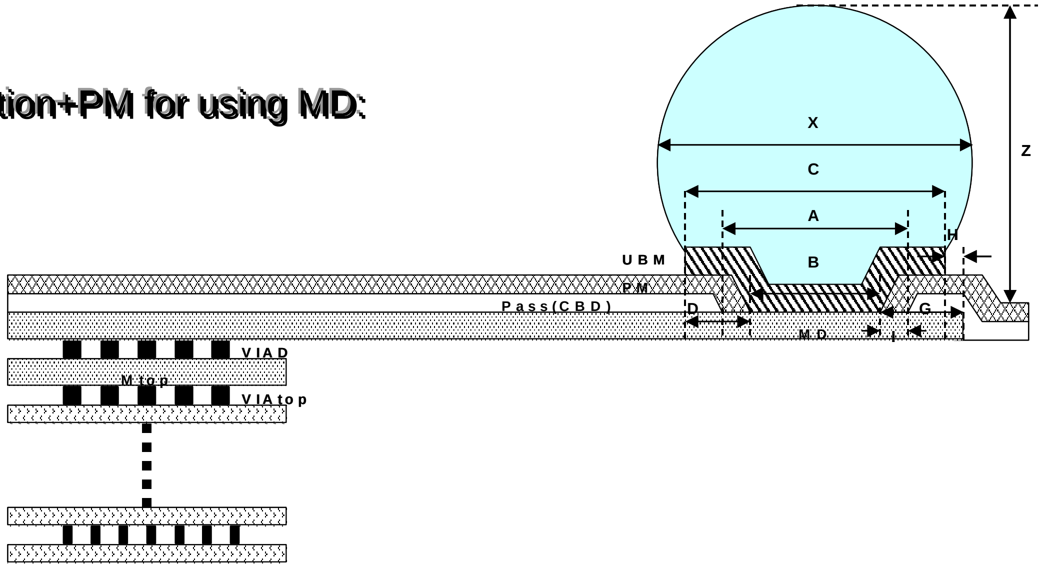
Top View:

Note: S, S2, P, Q, Q1, R are defined in UBM rules. Please refer to section 4.2.

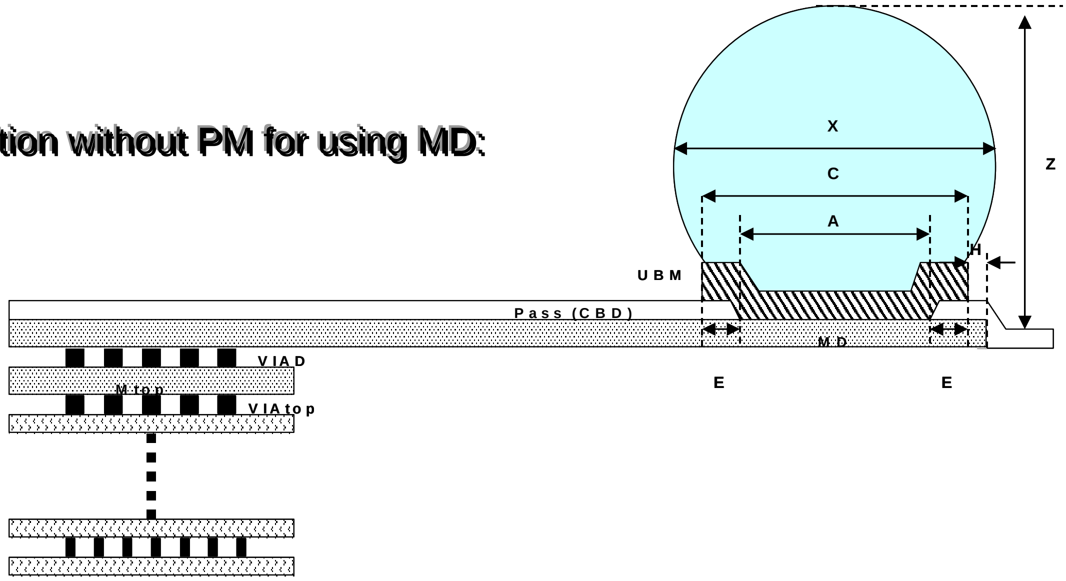
Cross Section:



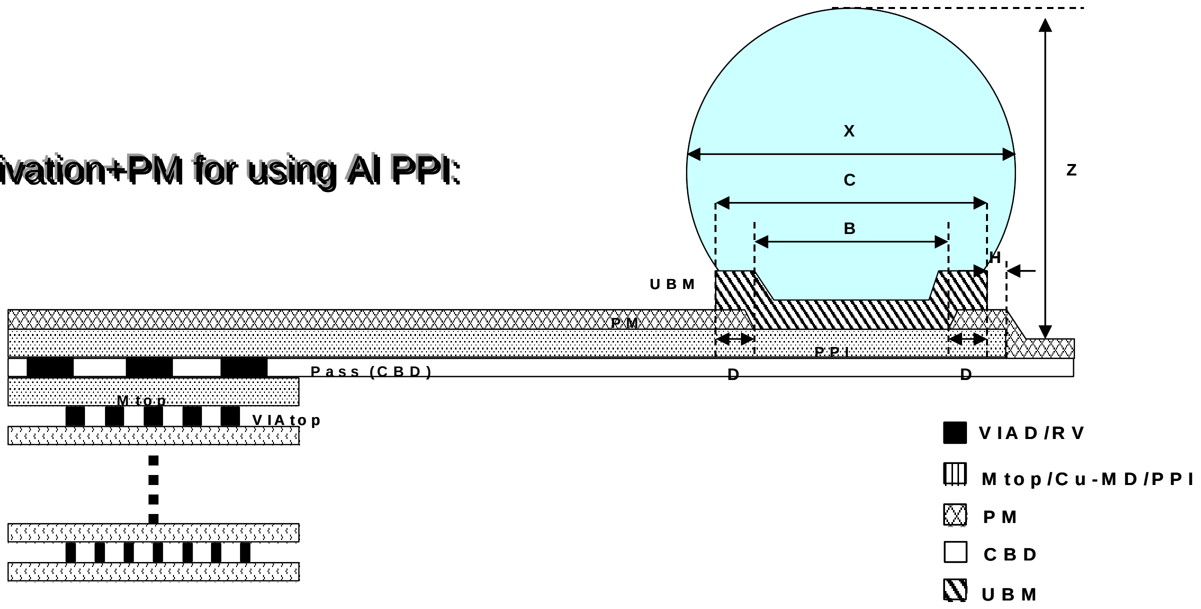
Passivation+PM for using MD:



Passivation without PM for using MD:



Passivation+PM for using Al PPI:

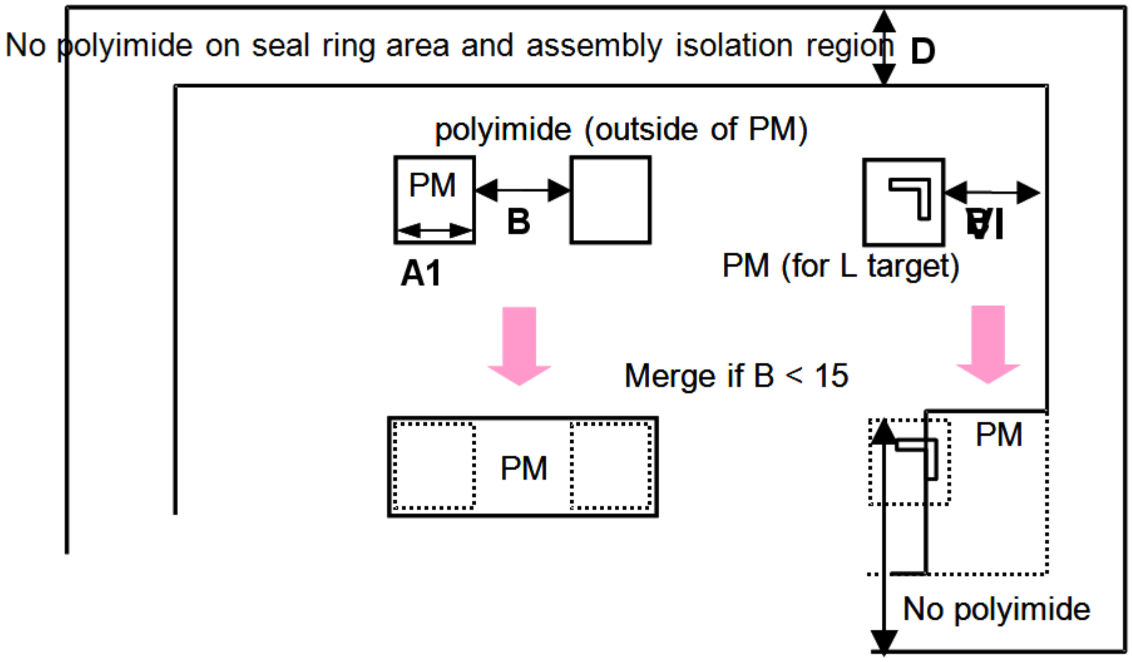


4.3.4 Polyimide (PM) Rules

- **Polyimide** is an option for C025 to C015 and its half node technologies. TSMC believes the standard passivation can cover all the package reliability requirements. If you have a special need for polyimide, please contact TSMC.
- The rules are based on TSMC process capability. If you do not use TSMC bumping service, please consult your bumping house regarding these rules, and the PM layer should be implemented at your bump house, not at TSMC. This is to prevent the unpredictable compatibility problem that might happen between PM and bumping process.
- PM is a drawn layer for flip chip design. It cannot be generated by logic operation. You need to define the CAD layer in the tape-out information.

Rule No.	Description	Label		Rule
PM.W.2	Width (Not interact with CBD region)	A1	≥	30
PM.S.1	Space of two PM, or space of PM to chip edge. Merge if space is less than 15µm.	B	≥	15
PM.R.3	Polyimide is prohibited over the seal ring area and assembly isolation region, LMARK and FW.	D		

PM:



4.3.5 Flip Chip Non-shrinkable Rules for the Half Node Technologies

- The layout dimensions of the following items need to be sized up before 90% shrinkage. For example, C018 → C016;

Rule No	Description		Rule
UBM.P.1	Pitch	≥	Table 4.3.3.1.a
UBM.S.1	Space to metal fuse protection ring	≥	55
UBM.S.2	Space to L target protection ring	≥	66
UBM.EN.1	Enclosure by chip edge	≥	88
UBM.EN.2	Enclosure by AP	≥	2.3
UBM.A.1®u	Area (without UBM in the chip center, for test inking requirement. Not necessary for inkless sorting)	≥	825*825
BP.W.1	Width of CBD under UBM area	≥	44
BP.W.2	Width of PM under UBM area (with Polyimide)	≥	33
BP.W.3	Width of UBM	≥	88
BP.EN.1	PM enclosure by UBM (with Polyimide)	≥	16.7
BP.EN.2	CBD enclosure by UBM (without Polyimide)	≥	11.2
BP.EN.4	PM enclosure by CBD (with Polyimide)	≥	11.2

*** Warning:** For design with a bump pitch below 160um (rule UBM.P.1, after shrinkage), please consult with your assembly house in advance. Make sure that your assembly house is able to provide such substrates and the associated service for your smaller bump pitch design.

4.3.6 Redistribution Metal Layout Rules

- Users can use the metal layer before passivation (MD), or after passivation (PPI) as the redistribution metal. The thickness of the metal layer are listed in the following table.

		Al Process	
Generation		C025~C015	
Redistribution Metal	Before/After passivation	Before	After
	Metal layer	MD	PPI
	Metal type	Al	Al
	Thickness (Å)	9.9K	20K

4.3.6.1 General guidelines of MD Layout Rules for Al Process (Redistribution Metal Before Passivation)

1. The standard brief process flow of redistribution interconnection using MD is documented in the following documents:
   - *CL015: T-015-LO-PF-011*
   - *CL025: T-025-LO-PF-011*
2. Please refer to the “I/O ESD Protection Guideline” section in the corresponding design rule manual.
   a. The recommended minimum BUS line width is 20 µm. (BUS line width = total width of all metal layers connecting with bump pad)
   b. The recommended maximum resistance of the BUS line from V_{dd} or V_{ss} pads to any I/O pad is 3Ω.
   c. The recommended minimum counts of VIAD connecting to a bump pad is 200 each for C025~C015.

4.3.6.2 MD and VIAD Rules

Rule No.	Description
MD.R.1	MD (RDL) is used to convert a product from a wire bond to a flip chip package. (DRC will flag MD without connecting to (CBD AND UBM) and design without (CBD OR UBM)

1. MD and VIAD must follow the Mtop and VIAtop rules of each logic design rule manual. Please refer to the following design rule document summary table.
2. VIAD is the via hole between two thick metals (Mtop and MD). The rule of VIAD enclosure by Mtop must follow the rule of Mtop enclosure of VIAtop.
3. For the seal ring, antenna, current density, stress relief, and metal slot rules of VIAD and MD, please refer to the corresponding rule items in the “Layout Rule Description” chapter of each logical design rule manual.

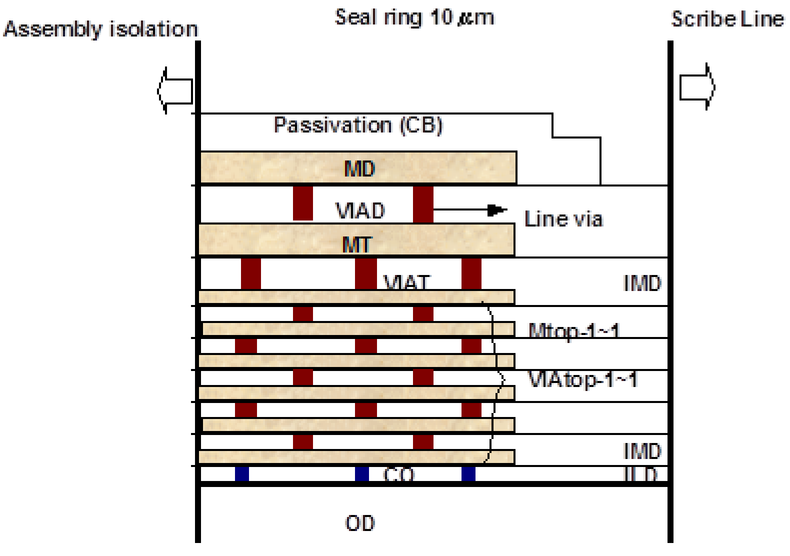
Technology code (Document No.)	Rule section in the "Layout Rule Description" chapter of each logical Design Rule Manual		
	VIAtop rule	**Mtop rule**	**Other metal rule**
C015 (*T-015-LO-DR-001*)	Via6 Rule (176)	Metal-7 Rule (187)	• Seal-Ring Rule • Antenna Effect Prevention Design Rule • Current Density Specification • Stress Relief Rule
C018 (*T-018-LO-DR-001*)	Via5 Rule (175)	Metal 6 Rule (186)	• Seal-Ring Rule • Antenna Effect Prevention Design Rule • Current Density Specification • Stress Relief Rule
C022 (*T-022-LO-DR-001*)	VIAx (x=1,2,3,4) Rule (178,179,173,174)	Metal-5 Rule (185)	• Seal-Ring Rule • Antenna Effect Prevention Design Rule • Current Density Specification • Stress Relief Rule
C025 (*T-025-LO-DR-001*)	VIAx (x=1,2,3,4) Rule (178,179,173,174)	Metal-5 Rule (185)	• Seal-Ring Rule • Antenna Effect Prevention Design Rule • Current Density Specification • Stress Relief Rule

4.3.6.3 Seal Ring Structure Using MD layer

- MD and VIAD layout within seal ring are the same as the Mtop and VIAtop layer.

C015~C025 seal ring cross-section:

- Please contact TSMC to get seal ring structure for PPI process.



4.3.7 RV And PPI Layout Rules for Al process (Redistribution Metal After passivation)

- RV is a CB VIA hole between PPI and Mtop. For design with redistribution metal after passivation (PPI), you should directly draw RV layout (CB VIA hole) and the passivation opening in the CBD layer (CAD layer number: 169).
- PPI process does not support WLCSP package.

4.3.7.1 RV Layout Rules (CB VIA holes)

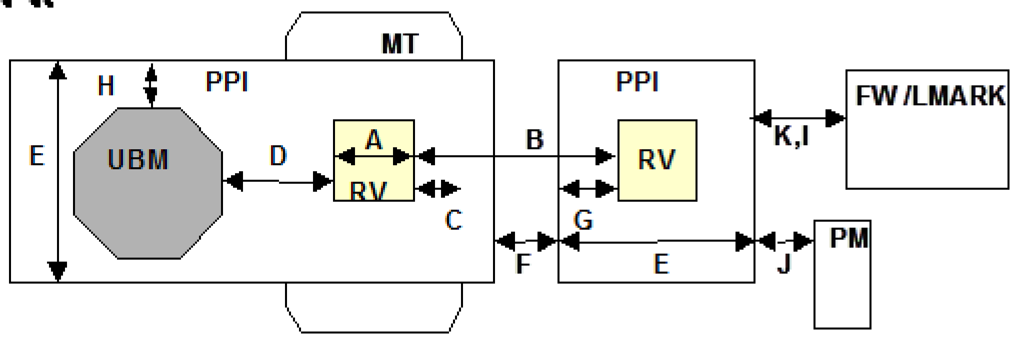
Rule No.	Description	Label		Rule
RV.W.2	Width (square) (maximum=minimum) (except seal ring area)	A	=	5
RV.S.1	Space	B	≥	3
RV.S.2	Space to UBM (Overlap is prohibited)	D	≥	0
RV.EN.2	Enclosure by Mtop (except seal ring area)	C	≥	2

4.3.7.2 PPI Layout Rules

If a PPI layer is not fabricated by the TSMC fab, customers must consult with the third-party assembly house about the related rules. However, if the PPI process has constraints on the layout of the passivation opening, customers must follow the Passivation Window CB rules before performing the PPI process.

Rule No.	Description	Label		Rule
PPI.W.1	Width	E	≥	10
PPI.S.1	Space	F	≥	5
PPI.S.2	Space to fuse window (FW)	K	≥	5
PPI.S.3	Space to L mark (LMARK)	I	≥	5
PPI.S.4	Space to PM. {PM AND PPI} is only allowed in the UBM area.	J	≥	5
PPI.EN.1	Enclosure of RV	G	≥	5
PPI.EN.2	Enclosure of UBM	H	≥	2
PPI.DN.1	Density (Total PPI metal area/ chip area) Dummy pattern is required when PPI density is less than 30%.		≥	30%

RV and PPI:



4.3.7.3 Seal Ring Structure

- PM is required in C015~C025 with PPI, but it is prohibited for polyimide to cover the seal ring and assembly isolation region. For the seal ring structure in PPI process, please refer to table 4.3.7.3.1
- Table 4.3.7.3.1 C015~C025 Seal Ring in PPI process reference Document

Technology	Refer to Document	Section
C025/C022	T-025-LO-DR-001	4.5.21 Seal Ring Rule
C018/C016	T-018-LO-DR-001	4.5.20 Seal Ring Rule
C015	T-015-LO-DR-001	4.5.20 Seal Ring Rule

Appendix A.1 Change from Each Document to Version 1.0

A.1.1 Merged from Design Rule Documents

Version 2.0 includes rules that were merged from the following Design Rule documents:

- *T-000-LO-DR-001: TSMC AL BOND PAD DESIGN RULE FOR WIRE BOND (Ver.2.3)*
- *T-000-LO-DR-004: TSMC CU BOND PAD DESIGN RULE FOR WIRE BOND (Ver.0.3)*
- *T-000-FC-DR-001: TSMC FLIP CHIP BOND PAD DESIGN RULE FOR AL PAD (Ver.2.1)*
- *T-000-FC-DR-002: TSMC FLIP CHIP BOND PAD DESIGN RULE FOR CU PAD (Ver.1.2)*
- *T-000-LO-DR-002: REDISTRIBUTION FOR SOLDER BUMP DESIGN RULE (Ver.0.4)*
- *T-000-FC-DR-003: TSMC POST PASSIVATION INTERCONNECTION (PPI) FOR SOLDER BUMP DESIGN RULE (Ver.0.2)*
- *T-000-LO-DR-005: TSMC CMOS LOGIC GENERAL PURPOSE COPPER INTERCONNECTION DESIGN RULE FOR FLIP CHIP (Ver.0.1)*
- *T-000-FC-DR-004: TSMC AL INTERCONNECT DESIGN RULE FOR FLIP CHIP (0.15/0.18/0.22/0.25UM) (Ver.0.1)*

A.1.2 Rule Changes from Pre-merged Document to this Version 1.0 Merged Document

T-000-LO-DR-001 (Ver.2.3)

Rule	Revision
CB	Added CB structure rules of CL022 (via and metal rules)
CB.EN.1	Revised CB.EN.1 from M1/M2 to M1~Mtop-3.
CB.R.1:	Changed the pattern of notch (50 µm pitch) and the length of probing area from 60 µm to 50 µm.
CB.W.1	Changed 48 µm to 43 µm for BGA and from 53 µm to 48 µm for non-BGA.
CB.W.3	• Changed M1/M2 to M1~Mtop-3 and 50. • Added 50 µm for 80 µm pitch staggered.
Double bonds	Added wording that double bonds are for QFP only.
Pitch (50 µm)	Merged 50 µm pitch rules (43 µm* 80 µm) with the CB rule for BGA.
PM.R.3	Revised PM.R.3 and removed the PM logical operation.
Staggered	Changed inner/out tier of staggered to inner/outer pad.
Tri-Tier	Changed first/second/third tier of Tri-Tier to inner/middle/outer pad.

T-000-LO-DR-004 (Ver.0.3)

Rule	Revision
CB.EN.1	Revised CB.EN.1 from M1/M2 to M2~Mtop-3.
CB.R.1/CB.W.3	Added rules.
CB.S.1	Add CB.S.1= 9 µm.for 50µm and 55µm pitch
CB.W.1	Add CB.W.1= 46 µm of 55µm pitch for all packages and = 41 µm of 50µm pitch for BGA only.
CBVIAT.EN.2	Revised CBVIAT.EN.2 from Mtop-1 to Mtop-1/Mtop (for CL013) and Mtop-2/Mtop-1/Mtop for (CN90).
Staggered	• Revised from inner/outer tier of staggered to inner/outer pad. • Added rules for pitch staggered (70 µm).
Tri-Tier	Added rules for pitch tri-tier (80 µm.

T-000-FC-DR-001 (Ver2.0)

Rule	Revision
BP.EN.3	Revised BP.EN.3 from 2 µm to 12 µm.
UBM.R.2	Revised the wording for UBM.R.2.
UBM.DN.1	Added rule.
UBM.EN.3	Revised UBM.EN.3 from 160 µm (from the center of the scribe lane) to 100 µm (from the chip edge).

T-000-FC-DR-002 (Ver.1.2)

Rule	Revision
BP.S.3/BP.S.4	Removed space between two AP (BP.S.3 and BP.S.4).
UBM.DN.1	Added rule.
UBM.EN.3	Revised UBM.EN.3 from 160 µm (from the center of the scribe lane) to 100 µm (from the chip edge).
UBM.R.2	Revised the wording for UBM.R.2.

T-000-LO-DR-002 (Ver.0.4)

Rule	Revision
Rule writing style	Revised the rule writing style to refer to the top VIA and the top metal rule of each generation's design rule.
VIAD.R.2 (Al)	Revised VIAD.R.2 (A1) from 400 to 200.

T-000-FC-DR-003 (Ver0.2)

Rule	Revision
Application process	Revised the application process from both Cu and Al processes to the Al process only.
CB.S.4	Added rule.

T-000-LO-DR-005 (Ver.1.0)

Rule	Revision
BP.R.7	Removed rule BP.R.7.
UBM.EN.1	Revised UBM.EN.1 from 120 µm (space to center of scribe lane) to 100 µm (enclosure by chip edge).

T-000-FC-DR-004 (Ver.0.1)

Rule	Revision
BP.R.7	Removed rule BP.R.7.
BP.R.9	Revised BP.R.9 from 170 µm (space to center of scribe lane) to 110 µm (space to chip edge).
UBM.EN.1	Revised UBM.EN.1 from 120 µm (space to center of scribe lane) to 100 µm (enclosure by chip edge).

Appendix A.2 Change from Version 1.0 to 1.1

New

	New rules (Version 1.0 to 1.1)	Description
1	New bump pad pitch rules (150~175μm) for Al and Cu process	Add new rules of UBM, CBD, PM, bump diameter, and bump height for 150 ~ 175μm pitch
2	BP.EN.4 for Al process	Add this rule of "PM enclosure by CBD region"
3	Mechanical and Thermal guidelines for FCBGA	Add new packaging guideline for CL013 and beyond technologies.
4	1.1 OVERVIEW	Add the wordings of " The pad pitch and packaging technology ..."
5	2.2 Flip Chip Guideline	Add " 3. For bump pad pitch selection:…"
6	2.5 Non-shrinkable Guidelines for Shrinkage Technologies	For easier use by the designer who wants to use the TSMC shrinkage technologies
7	PM.W.2	Add this rule for the polyimide window opening on non-CB region
8	4.7 Non-shrinkable Rule For Shrinkable Technologies	For easier use by the designer who wants to use the TSMC shrinkage technologies
9	Add the seal ring cross section on section 4.6.1 and section 4.6.2	Insert the MD/PPI layer into the seal ring cross-section

Modified

	Modified rules (Version 1.0 to 1.1)	Description
1	UBM.P.1	Modify from 175μm to 150μm
2	CB.W.2	Modify from "width" to "length"
3	BP.W.4	Add "… under UBM area"
4	The bump height and diameter in Table1 & Table2	Simplify the bump height and diameter information
5	BP.EN.2 for Al process	Add the description "without polyimide process only"
6	CB.EN.1	Remark this rule at section 4.4.2.
7	2.3 Redistribution Interconnection Guidelines 4. I/O ESD protection recommendations:	Modify "The minimum bus line width is 20μm." to The minimum total bus line width is 20μm.
8	1.1 OVERVIEW	Modify the wordings of "is only applied…" to "is verified by…"
9	Modify the description using "Mil" unit	Add "μm" in front of "Mil"
10	PM.W.1	Specify this rule using on PM opening on CB region.
11	PPI.S.3	Relax from 30μm to 5μm
12	Modify CB.EN.1 for Cu process	Relax this rule from 2μm to 1.5μm
13	BP.EN.2	Add"…. (Without Polyimide process only"
14	Modify "*" mark for DRC can't check	Change from "#" to "u"

Deleted

	Deleted rule (Version 1.0 to 1.1)	Description
1	60 μm pitch for single in-line Cu pad	Delete the rules of this pitch due to no impact on TSMC IO library.

Appendix A.3 Change from Version 1.1 to 1.2

New

	New rules (Version 1.1 to 1.2)	Description
1.	Add CLN65 relative wire bond and flip chip design rules	Wire bond pad structure need have new rules. Flip chip rules are same with CLN90 low-K
2	#CB.S.3 [t]	Space of the different pad geometries [(Single to Stagger), (Single to Tri-tier) or (Stagger to Tri-tier)]
3	CB.W.1/CB.S.1 for dual passivation	Relax CB.W.1/CB.S.1 to 44/6um(50um pitch) and 49/6um (60um pitch) for dual passivation only
4	60um pitch staggered rule	CB.W.1/CB.W.2/CB.S.1=50/66/10um .CB.W.1/CB.S.1 will be set as non-shrinkable.
5	UBM.S.4®	Recommended maximum space between two UBM
6	UBM.R.5	At least one bump has to be placed near, and as close as possible, to the chip corner.
7	UBM.R.5®	It's recommended not to place the IO bump pads on 2^{nd} and 3^{rd} row on bump array corners, but instead to put Vss, Vdd or dummy bump pads.
8	UBM.EN.1g	Recommended enclosure by chip edge (Maximum)
9	UBM.R.4g	• It is recommended not to put any bump on the top of SRAM area and the matching pairs. o The circuits should be located at a minimum distance of 60 μm from the bump pad's PM or CBD edge. o It's also recommended to consider UBM.S.4® at the same time. o If bump over SRAM area is needed, it's recommended to use the ultra-low alpha particle materials during the bump and assembly processes (solder bump, under-fill, pre-solder bump…) to avoid high Soft Error Rate (SER). o TSMC uses ultra-low alpha particle materials in the solder bump process. If a customer couldn't meet UBM.S.4® and UBM.R.4g at the same time, you can consult TSMC for layout suggestion.
10	#BP.S.1 [t]	PM space to AP-MD [PM on AP is prohibited, except UBM region and seal ring] (CL013 FSG with Al PPI)
11	BP.EN.8 [t]	PM enclosure by UBM (CL013 FSG with Al PPI)
12	BP.R.1 [t]	CB2 on AP-MD for interconnection is prohibited [except UBM and sealring]
13	CB-VD and AP-MD rules for Al PPI for CL013/CLN90 Cu process flip chip design	
	CB.W.1	Width (maximum =minimum) {Not inside seal ring}
	CB.S.1	Space
	CB.EN.1	Enclosure by MT {Not inside seal ring}
	CB.R.1	A 45-degree rotated CB-VD is prohibited
	AP.W.1	Width {Interconnection only} {Not inside UBM or seal ring}
	AP.S.1	Space
	AP.S.2	Space to FW
	AP.S.3	Space to LMARK
	AP.S.4	Space to CB2/PM
	AP.EN.1	Enclosure of CB-VD {Not inside seal ring}
	AP.DN.1	AP density across full chip
	AP.W.2® [U]	Recommended total width of BUS line [Connect with bump pad]
14	Add Sec. 2.1 wire bond guideline item 2	Pad Geometries selection
15	Add Sec. 2.1 wire bond guideline item 3	Pad pitch and size.
16	Add Sec. 2.2 Flip chip guideline item 6	Add new Polyimide guideline

Modified

	Modified rules (Version 1.1 to 1.2)	Description
1	#UBM.S.1 [t]	Space to metal fuse protection ring (change from 60 to 50um)
2	UBM.S.3 [t]	Space to L target (for Cu process only) (change from 90 to 80um)
3	Non-shrinkable rule of CB.W.2 of 55um pitch single	Change from 72u.6um to 66um.
4	Non-shrinkable of UBM.S.1	Change from 66 to 55um
5	Non-shrinkable of UBM.S.3	Change form 99 to 88um
6	Sec. 2.2 Flip chip guideline item 7	Modify this process sequence table and add Al PPI(AP-MD)
7	Sec. 2.2 Flip chip guideline item 9	Modify the alpha particle sensitive area and forbidden SRAM area ,except use ultra-low alpha particle material

Appendix A.4 Change from Version 1.2 to 1.3

	From Version 1.2 to Version 1.3			
	Rule	Sec. No.	Section Title	Revision Description
1.	Consolidate C015/C018 (T-000-CL-DR-001) and C013 CUP design rule (T-013-CL-DR-005)	2.1.2	Circuit Under Pad (CUP) Recommendations	Recommendations for CUP
		3.2	Special Recognition CAD Layers	Add WBDMY to cover CUP pad
		4.4	Pad Pitch Rules for Wire Bond	Add CUP relative descriptions in the current rules (CB.W.3/CB.S.2/CB.EN.1)
		4.6.2	Circuit Under Pad (CUP) Pad structure rules (for C015/C018 only)	Merge all rules from T-000-CL-DR-001
		4.7.2	Circuit Under Pad (CUP) Pad structure rules (for C013 only)	Merge all rules from T-013-CL-DR-005
2.	Add RV and AP-MD rule for C013/CN90/CN65 wire bond application	3.1	Mask Information, Process Sequence and CAD layers	Add the process sequence of AP-MD with dual passivation scheme on C013 and below technologies
		3.2	Special Recognition CAD Layers	Add RV, AP-MD and CB2 for AP RDL application
		4.4	Pad Pitch Rules for Wire Bond	Add AP-MD and CB2 relative descriptions in the current rules (CB.W.1/CB.W.2/CB.W.3/CB.S.1 /CB.S.2/CB.EN.1/CB.R.1)
		4.7	Wire Bond Pad Structure Rules for Cu Process	Add AP-MD and CB2 relative descriptions in the current rules (CB.R.5/CB.R.6/CBVIAx.R.3/ CBVIAx.R.3.1)
		4.8	RV And AP-MD Layout Rules for Wire Bond	Add RV and AP-MD rules for wire bond
		5.2	Polyimide Window (PM) Rules for Flip Chip	Add AP-MD and CB2 relative descriptions in PM.W.2
3.	Item 5: Required bumping testing flow	2.2	Recommendations for Flip Chip Application	Change “Suggested bumping and testing flow… to “Required bumping and testing flow…”
4.	Mask layer, Process Sequence Table	3.1	Mask Information, Process Sequence and CAD layers	Modify the table and separate the tables of “Al process,” “C013 single passivation,” C013/CN90 dual passivation,” and “CN65 dual passivation”
5.	60um pitch staggered pad rules	4.4	Pad Pitch Rules for Wire Bond	CN90/CN65 dual passivation and Al process single passsivation share the same rules and separated with C013.
6.	70um and 80um pitch staggered pad rules	4.4	Pad Pitch Rules for Wire Bond	Change to become the recommended rules
7.	PM.R.1/PM.R.2	4.5	Polyimide Window (PM) Rules for Wire Bond	Add “ (Al process only)”
8.	CBVIAx.R.1	4.6.1	Wire Bond Pad Structures Rules For Al Process	Modify the description to except the regions of CBVIAx.R.1.1
9.	CBVIAx.R.1.1	4.6.1	Wire Bond Pad Structures Rules For Al Process	Add this new rule for “Ratio of total exposure area of {VIA1~VIAtop-2 INSIDE CB} to CB area (In the inter row pad of staggered and the inter/middle row pad of Tri-tier)”
10.	CBVIAx.R.3	4.7.1	Wire Bond Pad Structures Rules For Al Process	Modify the description to except the regions of CBVIAx.R.3.1
11.	CBVIAx.R.3.1	4.7.1	Wire Bond Pad Structures Rules For Al Process	Add this new rule for “Ratio of total exposure area of {VIA1~VIAtop-2 INSIDE CB} to CB area (In the inter row pad of staggered and the inter/middle row pad of Tri-tier)”
12.	CN90(2XTM)/CN65(2XTM)/CN65(VIAy) rules	4.7.1	Wire Bond Pad Structures Rules For Al Process	Add new rules for CN90(2XTM)/CN65(2XTM)/ CN65 (VIAy) processes
13.	UBM.R.3	5.1	Under Bump Metallurgy (UBM) Rules	Add the required layers for the bump ball structure of different processes
14.	UBM.R.5®	5.1	Under Bump Metallurgy (UBM) Rules	Rename to UBM.R.5g
15.	UBM.R.6®	5.1	Under Bump Metallurgy (UBM) Rules	New guideline for the size ratio of UBM/Pre-Solder Bump SRO (=C/S) and BGA SRO/Board Pad (=T/O) = 1.0~1.1
16.	UBM.EN.1g	5.1	Under Bump Metallurgy (UBM) Rules	Rename to UBM.EN.1®
17.	BP.S.1	5.4	Bump Pad Structure Rules For Cu Process	Deleted this rule and replaced by AP.S.4
18.	BP.EN.7	5.4	Bump Pad Structure Rules For Cu Process	Relax the number from 15um to 2um.

	Rule	Sec. No.	Section Title	Revision Description
	From Version 1.2 to Version 1.3			
	Rule	**Sec. No.**	**Section Title**	**Revision Description**
19.	BP.EN.8	5.4	Bump Pad Structure Rules For Cu Process	Deleted this rule due to Single pass+PM scheme is prohibited for AP RDL process
20.	Rename all rules	5.5.2.1	RV Layout Rules (Passivation-1 VIA hole)	Rename CB.W.5/CB.S.3/CB.S.4/CB.EN.3 to RV.W.2/RV.S.1/RV.S.2/RV.EN.2
21.	RV.W.2	5.5.2.1	RV Layout Rules (Passivation-1 VIA hole)	Relax from >=25um to =5um
22.	RV.S.1	5.5.2.1	RV Layout Rules (Passivation-1 VIA hole)	Relax from >=15um to >=3um
23.	RV.S.2	5.5.2.1	RV Layout Rules (Passivation-1 VIA hole)	Relax from >=9um to >=0um
24.	Rename all rules	5.5.3.1	RV Layout Rules (Passivation-1 VIA hole)	Rename CB.W.1/CB.S.1/CB.EN.1/CB.R.1 to RV.W.1/RV.S.1/RV.EN.1/RV.R.1
25.	RV.S.2	5.5.3.1	RV Layout Rules (Passivation-1 VIA hole)	New rule
26.	RV.R.3	5.5.3.1	RV Layout Rules (Passivation-1 VIA hole)	New rule
27.	AP.W.1	5.5.3.2	AP-MD Layout Rules	Add CBD, CB2, and FW(AP) in the rule description
28.	AP.S.2/AP.S.3/AP.S.4	5.5.3.2	AP-MD Layout Rules	Add “ overlapping is prohibited in the rule description
29.	AP.EN.1	5.5.3.2	AP-MD Layout Rules	Modify CB-VD to RV
30.	AP.DN.1	5.5.3.2	AP-MD Layout Rules	Relax the maximum density from 60% to 70%
31.	A.R.10/A.R.11/A.R.12/ A.R.13	4.8.3 5.5.3	Layout Rules for RV and AP-MD	Add new rules Cu process.

Rule Number Mapping Table (from Version 1.2 to 1.3):

Consolidate from T-000-CL-DR-001 Ver1.2		
T-000-CL-DR-001	**Version1.3**	**Note**
CB.EN.1	CUPCB.EN.1	No rule change
CB.R.3	CUPCB.R.1	No rule change
CBVIAT.EN.2	CUPVIAT.EN.1	No rule change
CBVIAT.W.2	CUPVIAT.W.1	No rule change
CBVIAT.W.3	CUPVIAT.W.2	No rule change
CBVIAT.S.3	CUPVIAT.S.1	No rule change
CBVIAT.S.4	CUPVIAT.S.2	No rule change
CBVIAT.DN.1	CUPVIAT.DN.1	No rule change
CBVIAT.R.2	CUPVIAT.R.1	No rule change
Consolidate from T-013-CL-DR-005 Ver.1.2		
T-013-CL-DR-005	**Version1.3**	**Note**
CB.EN.3	CUPCB.EN.2	No rule change
CB.EN.4	CUPCB.EN.3	No rule change
CB.EN.5	CUPCB.EN.4	No rule change
CB.EN.6	CUPCB.EN.5	No rule change
CB.W.5	CUPCB.W.1	Change from =5um to >=5um
CB.R.3[u]	CUPCB.R.2[u]	No rule change
CB.R.4	CUPCB.R.3	No rule change
CB.R.5	CUPCB.R.4	No rule change
CB.R.6[u]	CUPCB.R.5[u]	Remove the constraint of “It’s prohibited to place CUP chip and non-CUP pad in the same chip”
CB.R.7	CUPCB.R.6	No rule change
CBVIAT.W.1	CUPVIAT.W.3	No rule change
CBVIAT.S.1	CUPVIAT.S.3	No rule change
CBVIAT.S.2	CUPVIAT.S.4	No rule change
CBVIAT.S.3	CUPVIAT.S.5	No rule change
CBVIAT.EN.1	CUPVIAT.EN.2	No rule change
CBVIAT.R.1	CUPVIAT.R.2	No rule change
CBVIAT.R.2	-	Remove
Rename the RV rules in section 5.5.2		
Version1.2	**Version1.3**	**Note**
CB.W.5**	RV.W.2	Change from 25um to 5um
CB.S.3**	RV.S.1	Change from15um to 3um
CB.S.4	RV.S.2	Change from 9um to 0um
CB.EN.3	RV.EN.2	
Rename the RV rules in section 5.5.3		
Version1.2	**Version1.3**	**Note**
CB.W.1	RV.W.1	No rule change
CB.S.1	RV.S.1	No rule change
-	RV.S.2	New rule
CB.EN.1	RV.EN.1	No rule change
CB.R.1	RV.R.1	No rule change
-	RV.R.3	New rule

Appendix A.5 Change from Version 1.3 to 1.4

From Version 1.3 to Version 1.4			
Rule	**Sec. No.**	**Section Title**	**Revision Description**
Consolidate N90/N65 CUP design rule (T-N90-CL-DR-009)	2.4	Special Recognition CAD Layers	Add WBDMY to cover CUP pad
	3.3.3	Pad Pitch Rules for Wire Bond	Add CUP relative descriptions in the current rules (CB.W.3/CB.S.2/CB.EN.1)
	3.3.4.2.2	Circuit Under Pad (CUP) Pad structure rules (for N65/N90)	Merge all rules from T-N90-CL-DR-009
A.R.10/A.R.11/A.R.12	3.3.7.3	Antenna Effect Prevention (A) Layout Rules for RV and AP-MD	0. Add antenna rules with core and I/O devices 1. Modify the A.R.12 parameter numbers
UBM.EN.1	4.2	Under Bump Metallurgy (UBM) Rules	Change rule number from 100um to 80um.
UBM.EN.1®	4.2	Under Bump Metallurgy (UBM) Rules	New rule for placing UBM uniform.
UBM.EN.2	4.2	Under Bump Metallurgy (UBM) Rules	New rule for placing UBM uniform.
UBM.EN.3	4.2	Under Bump Metallurgy (UBM) Rules	New rule for preventing iso bump.
UBM.EN.3®	4.2	Under Bump Metallurgy (UBM) Rules	New rule for preventing iso bump.
UBM.R.6®[u]	4.2	Under Bump Metallurgy (UBM) Rules	Change rule number from 1.0~1.1 to 0.95~1.05
UBM.EN.2	4.3.5	Flip Chip non-shrinkable Rules for the Half Node Technologies	Change rule number from 2.3 to 2.2
BP.EN.5	4.3.5	Flip Chip non-shrinkable Rules for the Half Node Technologies	Change rule number from 11.2 to 11

Rule Number Mapping Table (from Version 1.3 to 1.4):

Consolidate from T-N90-CL-DR-009 Ver.1.2		
T-N90-CL-DR-009	**Version1.4**	**Note**
CB.EN.3	CUPCB.EN.6	No rule change
CB.EN.4	CUPCB.EN.7	No rule change
CB.R.3[u]	CUPCB.EN.7[u]	No rule change
CB.R.6[u]	CUPCB.EN.8[u]	No rule change
CB.R.7	CUPCB.R.7	No rule change
CBVIAT.W.1	CUPVIAT.W.1	No rule change
CBVIAT.S.1	CUPVIAT.S.1	No rule change
CBVIAT.EN.1	CUPVIAT.EN.1	No rule change
CBVIAT.DN.1	CUPVIAT.DN.1	No rule change
CBVIAT.DN.2	CUPVIAT.DN.2	No rule change

Appendix A.6 Change from Version 1.4 to 1.5

From Version 1.4 to Version 1.5			
Rule	**Sec. No.**	**Section Title**	**Revision Description**
	Chap.1&2		Remove C013~N55 related wording
	Chap.3&4		Add re-qual warning for process change
	3.2.2	Reference Document	Add C018 CUP qualification report document no.
	3.3	WIRE BOND RULES FOR CU PROCESS (C013~N65)	Remove this section and merge this section into C013~N55 logic DRMs.
	4.1.1	General Layout Recommendations	Add table 4.1.1 for die size vs bump pitch & UBM width. Remove LF wording. Remove C013~N55 related wording
UBM.S.3	4.2	Under Bump Metallurgy (UBM) Rules	Remove “for Al process only”
UBM.S.3	4.2	Under Bump Metallurgy (UBM) Rules	Remove C013~N55 related rule
UBM.EN.2	4.2	Under Bump Metallurgy (UBM) Rules	Remove “for Al process only” & C013~N55 related rule
UBM.A.1 ᵘ	4.2	Under Bump Metallurgy (UBM) Rules	Change to recommendation rule
UBM.DN.1	4.2	Under Bump Metallurgy (UBM) Rules	Tighten from 2.5% to 10%
UBM.DN.1®	4.2	Under Bump Metallurgy (UBM) Rules	Replaced by updated UBM.DN.1
UBM.DN.2	4.2	Under Bump Metallurgy (UBM) Rules	Replaced by updated UBM.DN.1
UBM.DN.3	4.2	Under Bump Metallurgy (UBM) Rules	Remove “for the die size >= 169mm2”
UBM.DN.3®	4.2	Under Bump Metallurgy (UBM) Rules	Replaced by updated UBM.DN.3
UBM.R.1	4.2	Under Bump Metallurgy (UBM) Rules	Change to DRC checkable with adding wording “DRC only check the minimum vertexes of the equilateral polygon: 8.”
UBM.R.3	4.2	Under Bump Metallurgy (UBM) Rules	Remove C013~N55 related rule
UBM.R.5	4.2	Under Bump Metallurgy (UBM) Rules	Change to DRC checkable by redefinition
UBM.R.5g	4.2	Under Bump Metallurgy (UBM) Rules	Add “redundant”
UBM.R.7®	4.2	Under Bump Metallurgy (UBM) Rules	Add new recommendation rule
	4.2	Under Bump Metallurgy (UBM) Rules	Add figure 4.2.1 for updated UBM.R.5
	4.3.2	Reference Document	Add bump EM doc. no.
	4.3.3	Bump Pad Structure Rules (C025~C015)	Correct the table content for “After passivation(PPI)”
BP.EN.1	4.3.3	Bump Pad Structure Rules (C025~C015)	Add “PM not interact UBM is not allowed (except sealring, fuse, L-target).”
BP.EN.2	4.3.3	Bump Pad Structure Rules (C025~C015)	Add “CBD not interact UBM is not allowed (except sealring, fuse, L-target).”
BP.R.2®	4.3.3	Bump Pad Structure Rules (C025~C015)	Add new recommendation rule
	4.3.3	Bump Pad Structure Rules (C025~C015)	Add table 4.3.3.2 for UBM.R.7® & BP.R.2®
PM.R.3	4.3.4	Polyimide (PM) Rules	Add “LMARK and FW”
UBM.S.2	4.3.5	Flip Chip Non-shrinkable Rules for the Half Node Technologies	Remove “for Al process only”
UBM.A.1ᵘ	4.3.5	Flip Chip Non-shrinkable Rules for the Half Node Technologies	Change to recommendation rule
BP.W.1	4.3.5	Flip Chip Non-shrinkable Rules for the Half Node Technologies	Remove “for Al process only”
BP.W.2	4.3.5	Flip Chip Non-shrinkable Rules for the Half Node Technologies	Remove “for Al process only”
BP.W.3	4.3.5	Flip Chip Non-shrinkable Rules for the Half Node Technologies	Remove “for Al process only”
BP.EN.1	4.3.5	Flip Chip Non-shrinkable Rules for the Half Node Technologies	Remove “for Al process only”
BP.EN.2	4.3.5	Flip Chip Non-shrinkable Rules for the Half Node Technologies	Remove “for Al process only”
BP.EN.4	4.3.5	Flip Chip Non-shrinkable Rules for the Half Node Technologies	Remove “for Al process only”
	4.3.6.3	Seal Ring Structure Using MD layer	Add warning “· Please contact TSMC to get seal ring structure for PPI process.”
	4.3.7	RV And PPI Layout Rules for Al process (Redistribution Metal After passivation)	Add warning “· PPI process does not support WLCSP package.”
	4.3.7.3	Seal Ring Structure Using PPI layer	Remove “The seal ring structure is the same with C015~C025 DRM.” and add “Please contact TSMC to get seal ring structure for PPI process.”
	4.4	FLIP CHIP RULES FOR CU PROCESS (C013 ~N65)	Remove this section and merge this section into C013~N55 logic DRMs.
	Chap.5	MECHANICAL AND THERMAL RECOMMENDATIONS FOR FCBGA	Remove this section and merge this section into C013~N55 logic DRMs.

Appendix A.7 Change form Version 1.5 to 1.6

From Version 1.5 to Version 1.6			
Rule	**Sec. No.**	**Section Title**	**Revision Description**
	1.2	Applicable Technology for This Document	Update 40K thickness metal to C018/C016, and C025 CUP offering Table 1.2.2
	3.1.2	Recommendations for Circuit Under Pad	Deleted the sentence ''The design rules listed in this document are applied for C018~C015''.
	3.2.2	Reference Document	Added C025 qualification report
#CB.P.1[u]	3.2.3	Passivation Layer Open Rules	Revised wordings ''4 pads at the chip corner'' to 4 pitches at each chip corner.
	3.2.4.3	CUP(Circuit Under Pad)Pad Structure Rules	Add new section for C025 CUP
PM.R.3	3.2.5	Polyimide(PM) Rules for Wire Bond	Add wordings ''assembly isolation region'' in rule.
#CB.P.1[u]	3.2.6	Wire Bond Non-shrinkable Rules for the Half Node Technology	Revised wordings ''4 pads at the chip corner'' to 4 pitches at each chip corner.
	4.1.1	General Layout Recommendations	Statement 1 add wording ''only'' to prevent customers from being confused.
UBM.DN.3	4.2	Under Bump Metallurgy (UBM) Rules	Changed the rule indicate directly and added Figure 4.2.2
	4.3.3	Bump Pad Structure Rules(C025~C015)	Added Table 4.3.3.1.a and Table 4.3.3.2.a, C016 half node bump pad dimension for non-shrinkable rule
PM.R.3	4.3.4	Polyimide(PM) Rules	Add wordings ''assembly isolation region'' in rule.
	4.3.7.3	Seal Ring Structure	All seal ring structure in PPI process please refer to logic seal ring structure, and add a reference table 4.3.7.3

Appendix A.8 Change form Version 1.6 to 1.7

From Version 1.6 to Version 1.7			
Rule	**Sec. No.**	**Section Title**	**Revision Description**
	1.2	Applicable Technologies for This Document	Added latest description for new offering.
	1.2	Applicable Technologies for This Document	Updated Table 1.2.1 and table annotation
	1.2	Applicable Technologies for This Document	Updated Table 1.2.2 and table annotation
	2.2	Specific Geometries Used in Physical Design Rules	Added CUP phrases specific definitions in table 2.2.2
	3.1.1	General Layout Recommendations	Added CUP description in item 1.
	3.1.1	General Layout Recommendations	Added item 7
	3.1.1	General Layout Recommendations	Added item 8
	3.1.2	Recommendations for Circuit Under PAD (CUP)	Added CUP description in item 1.
	3.1.2	Recommendations for Circuit Under PAD (CUP)	Added CUP description in item 2
	3.1.2	Recommendations for Circuit Under PAD (CUP)	Added CUP description in item 4.
	3.1.2	Recommendations for Circuit Under PAD (CUP)	Added item 6
	3.1.2	Recommendations for Circuit Under PAD (CUP)	Added item 7
	3.1.2	Recommendations for Circuit Under PAD (CUP)	Added item 8
#CBWB.R.1[u]	3.2.3	Passivation Layer Open Rules	Added new rule
	3.2.4.2	Deformed Fully Stacking PAD (DFSP) structure for copper bond	Added new section
	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Added new section
DFSCB.EN.1	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy and revise from CUPCB.EN.1
DFSCB.R.1 [u]	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy and revise from CUPCB.R.1
DFSVIAT.EN.1	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy and revise from CUPVIAT.EN.1
DFSVIAT.W.1	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy and revise from CUP VIAT.W.1
DFSVIAT.W.2	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy and revise from CUPVIAT.W.2
DFSVIAT.S.1	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy and revise from CUPVIAT.S.1
DFSVIAT.S.2	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy and revise from CUPVIAT.S.2
DFSVIAT.DN.1	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy and revise from CUPVIAT.DN.1
DFSCB.EN.2	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy and revise from CB.EN.1
CB.R.2	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy from 3.2.4.1 CB.R.2
CBVIAx.W.1	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy from 3.2.4.1 CBVIAx.W.1
CBVIAx.S.1	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy from 3.2.4.1 CBVIAx.S.1
CBVIAx.S.2	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy from 3.2.4.1 CBVIAx.S.2
CBVIAx.EN.1	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy from 3.2.4.1 CBVIAx.EN.1
CBVIAx.R.1	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy from 3.2.4.1 CBVIAx.R.1
CBVIAx.R.1.1	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Copy from 3.2.4.1 CBVIAx.R.1.1
	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Added new section
DFSCB.EN.3	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Copy and revise from CUPCB.EN.2
DFSCB.R.2	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Copy and revise from CUPCB.R.2
DFSCB.R.3	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Added new rule
DFSVIAT.W.3	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Added new rule
DFSVIAT.EN.1	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Copy and revise from CUPVIAT.EN.1
DFSVIAT.DN.1	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Copy and revise from CUPVIAT.DN.1
DFSOBV.R.1	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Added new rule
DFSCB.EN.4	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Copy and revise from CB.EN.1
CB.R.2	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Copy from 3.2.4.1 CB.R.2
CBVIAx.W.1	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Copy from 3.2.4.1 CBVIAx.W.1
CBVIAx.S.1	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Copy from 3.2.4.1 CBVIAx.S.1
CBVIAx.S.2	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Copy from 3.2.4.1 CBVIAx.S.2
CBVIAx.EN.1	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Copy from 3.2.4.1 CBVIAx.EN.1
CBVIAx.R.1	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Copy from 3.2.4.1 CBVIAx.R.1
CBVIAx.R.1.1	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Copy from 3.2.4.1 CBVIAx.R.1.1
	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Added new section
DFSCB.EN.5	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Copy and revise from CUPCB.EN.2
DFSCB.R.2	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Copy and revise from CUPCB.R.2
DFSCB.R.3	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Added new rule
DFSVIAT.W.4	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Added new rule
DFSVIAT.S.3	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Added new rule
DFSVIAT.EN.2	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Copy and revise from CUPVIAT.EN.1
DFSVIAT.EN.3	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Added new rule
DFSVIAT.DN.1	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Added new rule
DFSMx.S.1	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Added new rule
DFSMx.S.2	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Added new rule
DFSOBV.R.1	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Added new rule
DFSCB.EN.4	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Copy and revise from CB.EN.1

From Version 1.6 to Version 1.7			
Rule	**Sec. No.**	**Section Title**	**Revision Description**
CB.R.2	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Copy from 3.2.4.1 CB.R.2
CBVIAx.W.1	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Copy from 3.2.4.1 CBVIAx.W.1
CBVIAx.S.1	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Copy from 3.2.4.1 CBVIAx.S.1
CBVIAx.S.2	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Copy from 3.2.4.1 CBVIAx.S.2
CBVIAx.EN.1	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Copy from 3.2.4.1 CBVIAx.EN.1
CBVIAx.R.1	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Copy from 3.2.4.1 CBVIAx.R.1
CBVIAx.R.1.1	3.2.4.2.3	Diamond via array deformation (3) C018 via array + OBV	Copy from 3.2.4.1 CBVIAx.R.1.1
CUPCB.EN.1	3.2.4.3	Triple Line(TrL) CUP Layout Rules	Rule description delete wording C018/C015
	3.2.4.4	One Big Via(OBV) CUP Layout Rules	Added description for section.
	3.2.4.4.1	C025 OBV Design Rules	Added new section
CUPCB.EN.2	3.2.4.4.1	C025 OBV Design Rules	Revised rule description
CUPCB.R.2	3.2.4.4.1	C025 OBV Design Rules	Revised rule description item 2 & 3
CUPCB.R.3	3.2.4.4.1	C025 OBV Design Rules	Added new rule
CUPVIAT.W.3	3.2.4.4.1	C025 OBV Design Rules	Added new rule
CUPVIAT.EN.1	3.2.4.4.1	C025 OBV Design Rules	Keep.
CUPVIAT.DN.1	3.2.4.4.1	C025 OBV Design Rules	Added new rule
OBV.R.1	3.2.4.4.1	C025 OBV Design Rules	Added new rule
	3.2.4.4.2	C018 OBV Design Rules	Added new section
CUPCB.EN.2	3.2.4.4.2	C018 OBV Design Rules	Added new rule
CUPCB.R.2	3.2.4.4.2	C018 OBV Design Rules	Added new rule
CUPCB.R.3	3.2.4.4.2	C018 OBV Design Rules	Added new rule
WBDMY.S.1	3.2.4.4.2	C018 OBV Design Rules	Added new rule
CUPVIAT.W.4	3.2.4.4.2	C018 OBV Design Rules	Added new rule
CUPVIAT.S.3	3.2.4.4.2	C018 OBV Design Rules	Added new rule
CUPVIAT.EN.2	3.2.4.4.2	C018 OBV Design Rules	Added new rule
CUPVIAT.EN.3	3.2.4.4.2	C018 OBV Design Rules	Added new rule
CUPVIAT.DN.1	3.2.4.4.2	C018 OBV Design Rules	Added new rule
CUPMx.S.1	3.2.4.4.2	C018 OBV Design Rules	Added new rule
CUPMx.S.2	3.2.4.4.2	C018 OBV Design Rules	Added new rule
OBV.R.1	3.2.4.4.2	C018 OBV Design Rules	Added new rule
	3.2.4.5	Single Metal (SM) CUP Pad structure Rules	Added all new section for CUP
CUPCB.EN.3	3.2.4.5	Single Metal (SM) CUP Pad structure Rules	Added new rule
CUPCB.R.4	3.2.4.5	Single Metal (SM) CUP Pad structure Rules	Added new rule

Appendix A.9 Change form Version 1.7 to 1.8

From Version 1.7 to Version 1.8			
Rule	**Sec. No.**	**Section Title**	**Revision Description**
	1.2	Applicable Technologies for This Document	Table1.2.2 C016 8K OBV open
	3.1.2	Recommendation for Circuit Under Pad(CUP)	Item 1 special metal scheme 1P2Mvia offer condition
	3.1.2	Recommendation for Circuit Under Pad(CUP)	Item 8 Table 3.1.2.2 0.18 Family half node C015, C152, C016, OBV related release.
TRL.R.1	3.2.4.2.1	Diamond via array deformation (1) via array + TrL	Added
DFSCB.R.3	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Delete
OBV.R.2	3.2.4.2.2	Diamond via array deformation (2) C025 via array + OBV	Added
DFSCB.R.3	3.2.4.2.3	Diamond via array deformation (3) C018/C016/C015 via array + OBV	Delete
DFSVIAT.S.3	3.2.4.2.3	Diamond via array deformation (3) C018/C016/C015 via array + OBV	Revised rule description
DFSMx.S.1	3.2.4.2.3	Diamond via array deformation (3) C018/C016/C015 via array + OBV	Delete
DFSMx.S.2	3.2.4.2.3	Diamond via array deformation (3) C018/C016/C015 via array + OBV	Delete
DFSCB.EN.4	3.2.4.2.3	Diamond via array deformation (3) C018/C016/C015 via array + OBV	Revised rule description
CBVIAx.EN.1	3.2.4.2.3	Diamond via array deformation (3) C018/C016/C015 via array + OBV	Revised rule description
OBV.R.1	3.2.4.2.3	Diamond via array deformation (3) C018/C016/C015 via array + OBV	Added
TRL.R.1	3.2.4.3	Triple Line (TrL) CUP Pad Structure Rules	Added
CUPCB.R.3	3.2.4.4.1	C025 OBV Design Rules	Delete
OBV.R.2	3.2.4.4.1	C025 OBV Design Rules	Added
CUPCB.R.3	3.2.4.4.2	C018/C016/C015 OBV Design Rules	Delete
CUPMx.S.1	3.2.4.4.2	C018/C016/C015 OBV Design Rules	Delete
CUPMx.S.2	3.2.4.4.2	C018/C016/C015 OBV Design Rules	Delete
OBV.R.2	3.2.4.4.2	C018/C016/C015 OBV Design Rules	Added
SM.R.1	3.2.4.5	Single Metal (SM) CUP Pad structure Rules	Added

Appendix A.10 Change form Version 1.8 to 1.9

From Version 1.8 to Version 1.9			
Rule	**Sec. No.**	**Section Title**	**Revision Description**
CB.R.2	3.2.4.1	Full Stacking Pad structure rules	Update description (flag CUP structure without WBDMY)
CB.R.3	3.2.4.1	Full Stacking Pad structure rules	Added new rule
DFSVIAT.EN.3	3.2.4.2.3	Diamond via array deformation (3) C018/C016/C0152/C015via array + OBV	Update description(delete the metal line 0.5um limitation)
CUPVIAT.EN.3	3.2.4.4.2	C018/C016/C0152/C015 OBV Design Rules	Update description(delete the metal line 0.5um limitation)
UBM.R.1	4.2	Under Bump Metallurgy (UBM) Rules	Update description
MD.R.1	4.3.6.2	MD and VIAD Rules	Added new rule

Appendix A.11 Change form Version 1.9 to 1.9_1

From Version 1.9 to Version 1.9_1			
Rule	**Sec. No.**	**Section Title**	**Revision Description**
	3.1.2	Recommendations for Circuit Under Pad (CUP)	Modify Table 3.1.2.2 for C015 30K UTM offering
CB.R.3	3.2.4.1	Full Stacking Pad structure rules	Modify description
CUPCB.R.3	3.2.4.3	Triple Line (TrL) CUP Pad Structure Rules	Add new rule
CUPCB.R.3	3.2.4.4	One Big Via (OBV) CUP Pad structure Rules	Add new rule
CUPCB.R.3.1	3.2.4.5	Single Metal (SM) CUP Pad structure Rules	Add new rule